

## **Luma Development Reference**

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# About Luma

Land Use Mapping for All (Luma) is an online mapping platform that integrates open-source spatial data and cloud-based computing for generating land use and land cover (LULC) maps. Luma is designed to accommodate a wide range of users through its user-friendly interface and guided steps, enabling anyone to easily produce custom LULC maps tailored to their needs.

This work is part of the **Evolving Participatory Information System for Nature-based Climate Solutions (Epistem)** initiative, which aims to develop an open-source landscape monitoring technology that can address multiple thematic requirements of diverse actors and stakeholders of nature-based climate solutions.

# Collaborative Development Guideline

## About the Three-Stream Approach

- **What is the three-stream approach?**

The three-stream approach is a workflow that separates brainstorming, design, and implementation into clear yet connected streams. Each stage of development has its own dedicated space, allowing for a structured and transparent decision-making and implementation process throughout the development cycle instead of mixing ideas, references, and implementations in scattered places.

- **Why the three-stream approach?**

The approach intends to solve a common problem in collaborative software development: ideas were discussed in one place, decisions were written somewhere else, and code evolved on its own, making it hard to tell what was current or why something changed. Someone working a week later or in a different time zone would often re-solve the same problem or unknowingly work against an outdated assumption.

The three-stream approach keeps context visible through [asynchronous communication](#), so work stays aligned even when people are not in the same place or working at the same time.

- **What are the three streams?**

- The “Braindump” (on Notion)

This is where ideas start. We use this Braindump to record all notes and discussions regarding future developments and improvements of the Luma Geospatial Engine. Nothing here is final and things are allowed to change or be discarded.

- The “Development Reference” (using Quarto)

This is where ideas become decisions. Content here is structured and stable enough for developers to refer to. It provides shared context so contributors can understand past decisions, even if they were not part of the original discussion.

- The “Code Implementation” (Luma Geospatial Engine Python package)

The Luma Geospatial Engine is the final implementation of the development, in the form of a Python-based package that powers the Luma platform.

Both the Development Reference and Code Implementation are hosted on GitHub for version control.

## Steps of the Three-Stream Approach

1. Start by writing your comments in the **Braindump** under the relevant section. Tag the person you want to discuss with so your input is visible and does not get missed.
  - If someone disagrees with your comment, add the topic to the **To-Settle** list. Use the page to write down the key arguments so they can be discussed efficiently in a meeting. Once the disagreement is resolved, move the topic to **Work in Progress**.
  - If there is no disagreement, you can add the topic directly to **Work in Progress**.
2. For agreed development work, create a new branch in **luma-dr** and document the proposed changes there.
3. Next, open a pull request in **luma-dr** from your new branch to the main branch. **Pull requests are reviewed during regular pull request meetings, where changes are either accepted or rejected.**
4. Once changes are merged into the main branch of **luma-dr**, contributors can begin implementing them in **luma-stack**. Implementation should always start from a new branch.
5. Submit a pull request on **luma-stack** to propose your changes for review at the next pull request meeting. If needed, also submit a pull request on **luma-dr** to update the documentation describing the changes.
6. During each pull request meeting, the team will:
  - Review and accept pull requests in **luma-stack** that implement previously approved changes
  - Update **luma-dr** based on approved pull requests
  - Review new pull requests in **luma-dr** proposed for future updates

### Tip

For more reference on the workflow (including a case example), refer to [this](#) slide presented during the Bootcamp.

## The “Braindump”

The Braindump serves as the central hub for refining ideas and discussions related to Luma’s development prior to their formal inclusion in the Development Reference. It contains organized pages that ensure alignment with the current Development Reference, along with sections dedicated to other discussion topics.

- **Structured pages for Luma Geospatial Engine development**

- The content of each page in this section reflects the latest version of the Development Reference (i.e., the `main` branch of `luma-dr`).
- Ideas and discussions can be provided by commenting on the related section of each page. **Tag the relevant person in the comment to ensure the comment is read by others.**
- The comments will be resolved in two cases:
  - \* No follow-up is needed after the discussion.
  - \* The follow-up has been listed in the **Work in Progress** or **To-Settle** section.

- **Structured pages for Luma User Interface development**

This section is a database dedicated to the implementation details of the Luma User Interface.

- **To-Settle**

- This section is dedicated to listing topics that arise from disagreements among multiple people in the comments on the **Structured pages for Luma Geospatial Engine development** and that require further discussion or a decision before updates are documented in the Development Reference.
- The disagreement should be briefly described inside the page.
- Choose an urgency level of the conflict to determine if the meeting should be scheduled as soon as possible.
- Once the conflict is settled, the issue can either be moved to be listed in the **Work in Progress** or dismissed.

- **Work in Progress**

- This section is dedicated for tracking ongoing tasks related to Luma Geospatial Engine updates.
- Each task goes through several stages of development according to the contribution workflow. The stage at which an issue is currently is noted in the **Completion status** column.

- **Unresolved issues from Pull Request approvals**

This section records issues or feedback that emerged during the pull request review process and need additional work or clarification before approval.

- **Meeting Notes**

A collection of notes, summaries, and key decisions from meetings related to Luma's development activities.

- **Sketchbooks**

A space for “sketching out” ideas or conceptual designs before formal documentation. This section is intentionally unstructured to encourage open exploration.

## Version Control

Version control is a core component of the Luma ecosystem and is managed through GitHub, which serves as the central platform for tracking both Quarto based Development Reference and Luma Geospatial Engine code implementation. GitHub provides a transparent and traceable environment for managing changes to code and documentation over time. Contributors create a dedicated branch for each feature, fix, or documentation update. Once the work is complete, changes are submitted to the main branch through a pull request. Each pull-request is reviewed and approved before merging to ensure quality control and alignment between the development reference and the geospatial engine.

## Repositories

1. There are two main repositories currently operating:
  - [luma-stack](#): This repository is used to host Luma Geospatial Engine Python package. To contribute to this repository, please refer to [README.md](#) in this repository for detailed instructions on setting up a developer environment and configuring your local machine.
  - [luma-dr](#): This repository hosts Luma Development Reference. Contributions follow this workflow:
    1. Clone the repository to your local machine.
    2. Create a new branch for your changes.
    3. Edit the relevant `.qmd` files to implement your updates.
    4. Document your modifications in the Braindump under the “Work in Progress” section.
    5. Test your changes by rendering the Quarto book locally to ensure correctness.
    6. Commit and push your changes to your branch.

7. Submit a pull request for review. All changes must be merged via pull request.
  8. Proposed changes will be discussed, and once approved, merged into the main branch. Your feature branch will then be deleted.
2. If a new repository is needed (e.g., for an experimental module, plugin, or related service), please consult with an Epistem admin before proceeding.

## Branches

1. Every repository should at least have one **main** branch. This branch represents the most up-to-date and stable version of the project.
2. A **branch protection** is targeted for the **main** branch. This means contributors cannot directly make changes to the **main** branch. Instead, all updates must go through a pull request (PR) process to ensure the code is reviewed and tested before merging. The branch protection rules are as the following:
  - Require a pull-request before merging (please refer to the **Pull-Requests (PRs)** section in this page for more details)
  - Require at least one approval before merging.
3. Each contributor should create a personal working branch from the **main** branch when developing new features or fixes. Once the work is complete, open a pull request to propose merging those changes into the **main** branch, unless there are circumstances where new branch can be created from another feature branch if the work is closely related.
4. Branch naming convention to specify the specific type of a contribution should be further discussed (e.g., **feature/**, **bugfix/**)

## Pull-Requests (PRs)

1. All contributions should be submitted through a pull request (PR) from your own branch. A PR is a formal request for your changes to be reviewed and merged into the **main** branch.
2. Keep a PR focused on a specific topic. This specific topic should also correspond to a task in the **Work in Progress** list in Notion. To help identify the task, name the PR title accordingly with what is written in the **Work in Progress** task list.
3. *Only for **luma-stack** and **luma-dr** repository:* when an update in one repository depends on or affects the other, link the related PRs in their descriptions. For example, if a **luma-stack** update requires changes in **luma-dr**, include a link to the corresponding PR in each repository to maintain traceability.

4. As required by the branch protection rules, every PR must receive at least one approval before merging. This approval should come from an Epistem admin. You may also request additional reviewers for feedback.
5. Reviewers should check for correctness, clarity, performance, and proper documentation. Use GitHub's review options:
  - Comment: suggest improvements
  - Request changes: block merge until issues are resolved
  - Approve: accept the PR as ready to merge
6. The PR title should reflect the main change made. This title will be used as the commit message when the PR is merged into the main branch.

## Contributor's Roles

1. Only members of the Epistem consortium are authorized contributors to these repositories.
2. There are three different roles set in the [epistem-io](#) organization, each with different levels of access:
  - Admin: has full control of repositories (e.g., create/delete repositories, manage access, merge pull requests).
  - Manage: can review and approve contributions, manage issues, and organize repository settings.
  - Write: can create branches, push code, and submit pull requests for review.

Contributors should know their assigned role before making any changes to ensure proper permissions are in place.

## Security and Secrets

**Never commit secrets** such as passwords, access tokens, or API keys. Add `.gitignore` file to prevent private files from being accidentally uploaded.

## Licensing

To be developed. A standardized licensing framework will be provided to ensure consistent open-source and internal usage across repositories.

## Releases

To be developed. The release process defines how stable versions of Luma tools and documentation are packaged, tagged, and shared. Releases ensure that users and collaborators can always access a verified version of the software or documentation.

## The Development Reference

The development reference is dedicated as a platform for mainly two purposes:

1. **Formalized the decision made in the Braindump.** All discussions' results that were made in the Braindump are documented accordingly, so that all developments made for the Code Implementation are well-tracked.
2. **Facilitate the development of Luma's technical documentation for end-users.** With the use of the development reference as a structured documentation of Luma's development, this documentation can also be a starting point to create Luma's technical documentation for end-users in the future, with reduced effort.

## Writing Guidelines for the Development Reference

- The development reference includes several pages, each serving a different thematic purpose in Luma technology
- Each of the pages is written in the following structure:
  1. **Title of the page**
  2. **Input:** a list of input data needed for the related process in the related page. The inputs are divided into:
    - User's Input: list of inputs provided by Luma's users.
    - Input from External Sources: list of inputs received from sources other than Luma's user.
    - Input from Other Modules: list of inputs from other modules in Luma.
  3. **Output:** list of expected output from the process defined in the page.
  4. **Process:** a detailed step-by-step the execution logic of the functionality described on the page. The process should be defined into the following sub-headings:
    - **Step:** a step represents a distinct change in system state or responsibility. A new step must be introduced when:
      - \* ownership shifts (e.g. from user action to system execution, or between modules), or
      - \* a new input set is consumed, or
      - \* an intermediate output is produced that is required by subsequent logic.

- **Sub-step:** a sub-step represents an atomic action required to complete a step. Sub-steps must not introduce new inputs or outputs beyond those defined for the parent step.
  - \* **Luma User Journey:** the description of the processes that occur in the user interface. This part is equivalent to the “User Journey” in the Visio workflow diagram.
    1. An error notification should be written in the `callout-important` block type with the title “Error Handling Notification”.
    2. A success notification should be written in the `callout-note` block type with the title “Success Notification”.
  - \* **Luma Geospatial Engine:** the description of the processes that occur in the back-end part. This part is equivalent to the “System Response” in the Visio workflow diagram.
    1. The defined function of the related process in the sub-step should be written in the `callout-tip` block type with the title “Related Function”.
- Write the Development Reference using clear and easy-to-understand wording that can be understood by audiences with different levels of technical expertise. Avoid technical jargon where possible while still ensuring that all critical elements of the process are clearly and sufficiently described.

**! Important**

When writing descriptions of a process, make sure to include all the crucial steps that need to be taken in the related process.

## Python Method Writing Convention

To be developed

# Testing Guideline for Developers

# **Part I**

## **General Development Settings**

# Python Environment Setup

## 1. Overview

Luma stands for Land Use Mapping for All. The geospatial functionalities that powers Luma is called Luma-GE (Luma Geospatial Engine). Luma-GE is shipped as a Python package because it needs to run in more than one context. The same geospatial engine must be importable by an interface layer (for example, a Streamlit-based UI/UX) and also be usable directly from a notebook for development, debugging, and method exploration. Packaging turns “a folder of scripts” into a reliable, versioned engine with a stable import path (`import luma_ge`), repeatable installs across laptops and CI, and a single place to declare runtime requirements.

Luma-GE uses `pyproject.toml` as the source of truth for packaging and installation. This file must live at the repository root. Modern Python tooling (including `pip`) reads `pyproject.toml` to understand the package identity, its required Python version, and the dependencies that must be installed alongside the engine. In practical terms, the `[project]` section is where we define the package name and version, set `requires-python`, and list the dependencies needed by the functions and methods provided by Luma-GE.

### Warning

Dependencies will be kept **parsimonious**. Only add a dependency when a function or method genuinely needs it. This keeps installs faster, reduces conflicts, and avoids dragging heavy geospatial binaries into environments that don't need them.

---

## 2. Goals

The goal is a developer setup that works the same way on different laptops and in CI. Installs should be standard and boring, and a new contributor should be able to get a working environment quickly without any one-off hacks.

---

### 3. Prerequisites

You need Git installed because `pip` installs from a Git URL by calling `git` under the hood. You also need a Python environment manager. For geospatial stacks, we recommend Miniforge or Mambaforge (Conda-compatible and lightweight) because many geospatial dependencies are easier to install as pre-built binaries than to compile locally.

To confirm Git and Conda/Mamba are available in your shell, run:

```
git --version  
conda --version
```

If you are on Windows, avoid adding Python or Conda to your system PATH unless you know exactly why you are doing it. PATH conflicts are a common source of broken environments.

---

### 4. Output

A working development environment that can run Luma-GE via an interface or via a notebook, and can also build package artefacts when needed.

---

### 5. Process

#### Step 1 - Create an isolated environment

Create a dedicated environment for Luma so your system Python stays untouched and dependency conflicts are contained.

```
mamba create -n luma python=3.11  
mamba activate luma
```

If you use `conda` instead of `mamba`, the commands are the same (just slower).

## Step 2 - Install Luma-GE from GitHub

If you want a quick install of the latest code without cloning the repo, install directly from GitHub. By default we install from the main branch.

```
python -m pip install "luma_ge @ git+https://github.com/epistem-io/luma-stack.git"
```

If you need a specific branch (for example, to test a feature branch or reproduce a bug), pin it explicitly:

```
python -m pip install "luma_ge @ git+https://github.com/epistem-io/luma-stack.git@<branch-name>"
```

### ! Important

Installing from a Git URL requires `git` to be installed and available on your PATH.

## Step 3 - Quick verification

Confirm the package imports correctly.

```
python -c "import luma_ge; print('luma_ge import OK')"
```

If you want a concrete example of how Luma-GE functions and methods are used in practice (outside the interface layer), review the implementation notebook in the repository:

`notebooks/Module_implementation.ipynb`

[https://github.com/epistem-io/luma-stack/blob/main/notebooks/Module\\_implementation.ipynb](https://github.com/epistem-io/luma-stack/blob/main/notebooks/Module_implementation.ipynb)

## Step 4 - Tests (coming soon)

### ⚠ Warning

#### Coming soon

A PyTest suite is planned but not implemented yet.

When it lands, this section will document the default test command (`pytest -q`), where tests live (`tests/`), naming (`test_*.py`), and recommended patterns (fixtures, `monkeypatch`, and mocking for cloud APIs).

## 6. Change management (three-stream approach)

Python environment changes are high-impact when they change the supported Python version, alter core dependency groups (especially geospatial binaries), modify install commands or extras, or change CI behaviour. These changes should follow the three-stream workflow (Notion → luma-dr → luma-stack) so that the documentation and implementation stay in lock-step.

---

## 7. References

- UBC Geography (2024) *Getting Started with Conda, Virtual Environments, and Python*. Available at: <https://ubc-geography.github.io/computing-resources/development-environments/conda-getting-started.html>
- pyOpenSci (2025) *Python Package Structure & Layout*. pyOpenSci Python Package Guide. Available at: <https://www.pyopensci.org/python-package-guide/package-structure-code/python-package-structure.html>
- PyPA (2025) *VCS Support*. Available at: <https://pip.pypa.io/en/stable/topics/vcs-support/>
- PyPA (2025) *Writing your pyproject.toml*. Python Packaging User Guide. Available at: <https://packaging.python.org/en/latest/guides/writing-pyproject-toml/>

# Earth Engine Initialization and Authentication

## 1. Overview

Luma-GE uses the Google Earth Engine (GEE) Python API for cloud-based geospatial processing. Before any module can call Earth Engine, the runtime must be authorised.

In practice, “authorisation” is two things glued together:

First, you authenticate as an identity (a human Google account, or a service account).

Second, Earth Engine needs a Google Cloud project context so quotas and usage accounting are applied to the right place. Recent Earth Engine Python client changes have made this project context effectively mandatory for common authentication modes. If Earth Engine cannot determine a “quota project”, requests will fail.

Luma-GE currently supports two working authorisation options, each with clear trade-offs:

1. Manual authentication (interactive OAuth) is best for local development and workflows tied to a human identity. It is interactive (browser + copy/paste code) and is not suitable for unattended deployments.
2. Service account authentication is the default for deployments and CI. It is non-interactive and repeatable, but it has no option for exporting to a user’s personal Google Drive.

## 2. Prerequisites

To use Earth Engine from Python, you need:

A Google account with Earth Engine access, and a Google Cloud project that is registered for Earth Engine and has the Earth Engine API enabled.

If you are using service account mode, you also need a service account created in that Cloud project and a JSON private key for it.

For service accounts, ensure the account has the correct IAM permissions on the project. In practice, the Earth Engine docs point you at Earth Engine resource roles, and some auth modes also require permissions related to Service Usage.

### 3. Output

An authorised session allowing Luma-GE functions to execute Earth Engine computations, traceable to a specific Google Cloud project.

### 4. Steps

#### 4a. Manual authentication (interactive OAuth)

Manual authentication is the simplest path for developers working locally. It authenticates you as your own Google account. That matters when you need behaviour tied to a human or organisation identity, such as exporting to your Google Drive.

In practice, you run an auth function, a browser window opens, you sign in with a Google account that has Earth Engine access, and you paste an authorisation code back into the terminal (or a notebook cell). After that, initialise Earth Engine with an explicit project ID so usage is attributed correctly.

In Luma-GE, this is implemented in `luma_ge/ee_config.py:authenticate_manually()`, which calls `ee.Authenticate()` followed by `ee.Initialize(project='YOUR_PROJECT_ID')`.

#### Warning

Luma-GE does not support running manual authentication inside a Streamlit interface. Earth Engine's interactive flow expects terminal/browser control that a hosted app generally does not have.

#### 4b. Service account authentication (deployment and CI)

A service account is a non-human Google identity intended for machine-to-machine access. This is the recommended mode for deployments, containers, scheduled jobs, and CI pipelines.

To create and download a service account key, use Google Cloud IAM: create a service account under your project, then create a JSON key for it. Treat the JSON key like a password. If it leaks, rotate it.

In Luma-GE, service account auth is implemented in `luma_ge/ee_config.py:initialize_with_service_account_key()`. The implementation loads the JSON key, sets `GOOGLE_APPLICATION_CREDENTIALS`, and initialises Earth Engine. If you do not pass a project ID explicitly, it attempts to infer it from the JSON (`project_id`).

Where should the key live? Anywhere except Git.

For local development, keep the JSON file outside the repo and point to it using `GOOGLE_APPLICATION_CREDENTIALS`, or store it in a local `auth/` directory that is excluded by `.gitignore`.

For deployments, inject the key via environment variables or a secrets manager. Luma-GE's credential discovery helper (`luma_ge/__init__.py:_find_service_account_file()`) searches in this priority order: `GOOGLE_SERVICE_ACCOUNT_JSON_B64`, then `GOOGLE_SERVICE_ACCOUNT_JSON`, then `GOOGLE_APPLICATION_CREDENTIALS`, before looking for local `auth/` folders and container fallbacks.

### ! Important

Never commit a service account key to GitHub. Public repos are scraped, history is hard to erase, and forks/caches persist. If a key is exposed, rotate it immediately.

Exports are the main bottleneck with service accounts. Earth Engine supports Drive exports via `Export.image.toDrive`, but Drive exports target the authenticated identity's Drive.

A recent Google Workspace policy change adds a very specific sharp edge. From 15 April 2025, newly created Google Cloud IAM service accounts no longer receive the default Google Workspace storage and cannot own Drive items. In practical terms, a service account created after that date will generally fail to export to "My Drive", because there is no Drive storage space for it to own the exported file.

For service accounts, the recommended pathways are exporting to Google Cloud Storage (GCS) or to an Earth Engine Asset, then sharing/copying results as needed. If you must land results in Drive, a more robust pattern is exporting into a Shared Drive, where the Shared Drive owns the file rather than the service account.

If you want a direct download link for quick debugging or open access downloads (without user login), `ee.Image.getDownloadURL()` is available but intentionally limited (maximum request size 32 MB and maximum grid dimension 10,000 pixels). It will not work for large rasters.

## 5. Troubleshooting

Most authorisation failures fall into a handful of predictable buckets.

### 5.1 Authentication fails

If Earth Engine says you are not authenticated, you either have no cached user credentials (manual mode) or the service account key was not found/loaded (service account mode). In Luma-GE, `get_auth_status()` is the quickest sanity check because it performs a trivial request to confirm the session can execute.

If manual authentication suddenly starts failing after a library update (especially in notebook-style flows), check the Earth Engine Python client library release notes. A common error message asks you to grant `roles/serviceusage.serviceUsageConsumer` (or the underlying `serviceusage.services.use` permission) on the relevant project.

If a service account works on one developer laptop but fails on another, it is usually one of these: the key file path is wrong or unreadable; the project is not registered for Earth Engine access; the Earth Engine API is not enabled; the service account does not have the required IAM roles; or permissions have not yet propagated.

## 5.2 Quota / project errors

If you see an error about a quota project not being set, Earth Engine cannot determine which Cloud project should be charged for requests. The Earth Engine guide recommends setting the project explicitly.

Ensure that (a) the project is registered for Earth Engine access, (b) the Earth Engine API is enabled in Cloud Console, and (c) you initialise with an explicit project in code: `ee.Initialize(project='YOUR_PROJECT_ID')`.

If you are using Application Default Credentials, the Earth Engine guide also suggests running `earthengine set_project YOUR_PROJECT_ID` or `gcloud auth application-default set-quota-project YOUR_PROJECT_ID`.

## 5.3 Exports fail

Export failures are a common trap. `ee.Image.getDownloadURL()` is intentionally limited to small downloads (maximum request size 32 MB and maximum grid dimension 10,000). If you exceed those limits, switch to batch exports.

For service accounts, it is suggested exporting to Cloud Storage (`Export.image.toCloudStorage`) or to an Earth Engine Asset. For large exports into a user's personal Drive, use manual authentication.

To understand limits and constraints applied to a project (task concurrency, storage, request rates, and other quotas), use the Earth Engine quotas and monitoring documentation, plus Cloud Console quota and monitoring pages.

## 6. References

Google for Developers (2025) *Authentication and Initialization*. Google Earth Engine Guides. Available at: <https://developers.google.com/earth-engine/guides/auth>

Google for Developers (2025) *Earth Engine access*. Google Earth Engine Guides. Available at: <https://developers.google.com/earth-engine/guides/access>

Google for Developers (2025) *Service Accounts*. Google Earth Engine Guides. Available at: [https://developers.google.com/earth-engine/guides/service\\_account](https://developers.google.com/earth-engine/guides/service_account)

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Google Cloud (2025) *Set the quota project*. Google Cloud documentation. Available at: <https://docs.cloud.google.com/docs/quotas/set-quota-project>

## Helpers

# Overview

The helper module is design to handle features or operations that will be utilized by several modules. Currently, the module handle data output specially raster based data. This operations is utilized by module 1, 5, and 6. Furthermore, there are several options for data output namely, direct download, google drive download, and cloud storage export. The complex export options are wrapped by helpers module for ease integration with front end code

# Input

| Name of Input                 | Input Type    | Details                                                                                 |
|-------------------------------|---------------|-----------------------------------------------------------------------------------------|
| Earth Engine image (ee.Image) | Module output | magery composite (module 1), predictors (module 5), and classification image (module 6) |
| Export parameters             | User's input  | File name, destination, projection, and scale (pixel size)                              |

# Output

Raster based data (GeoTiff)

# Process

## Earth Engine Export Utilities

### Export Parameters

Prior to export any earth engine image, the user must define several parameters required for export

#### Luma User Journey

1. The user select which export option they are going to use
2. The user define the required parameters for export. These parameters are as follows:
  1. File name (default file name is provided according to each module output)
  2. Spatial resolution (default to 30 m, since Landsat mission is the only one supported)
  3. Coordinate Reference System (CRS). Use WGS 1984 (EPSG:4326) as the default value. If the user wish to use different CRS, they must specified the EPSG code (i.e *'EPSG:4326'*, *'ESRI:54009'*, or *'SR-ORG:6864'*)
  4. For module 5, the user select which predictors to be exported. The predictor can be selected one by one, stacked of predictors, or by category

#### Feature Note

Current version of Luma GE only support direct download, which limits the data size to 32 MB. Google Cloud Storage (GCS) and google drive download are not supported in this phase.

#### Luma Geospatial Engine

1. The system creates a predefined default names according to the module and dataset exported.
2. Extract the geometry from the selected AOI, as prerequisites for export
3. Validate scale and CRS and raise a helpful error if invalid scale or CRS is detected

#### 💡 Related Function

`helper.EE_Export_Manager.sanitize_export_name()`: Standardize user input file name according to the system need. Replace invalid characters for GCS, Drive, and local file system

`helper.EE_Export_Manager.extract_geometry()`: Extract the geometry from `ee.featureCollection(AOI)` for export purpose

`helper.EE_Export_Manager.validate_scale()`: Validate user input pixel size value, design to avoid negative value what might cause the system to break. Default value use 30 m

`helper.EE_Export_Manager.validate_crs()` : Perform semantic validation for user input CRS code. The CRS content itself is not validated. Default values is WGS 1984 (EPSG:4326)

`helper.EE_Export_Manager.validate_image()` : Validate if the object pass to export manager is an `ee.image`

`helper.EE_Export_Manager.validate_export_params()` : validate the entire export parameters, raise helpful errors if detected

## Export Process

### Direct Download

Luma Geospatial Engine

1. The direct download function in Luma-GE used Earth Engine's `ee.Image.getDownloadURL` function, which generate a link for small chunks of image data in GeoTIFF or NumPy format. Maximum request size is 32 MB, maximum grid dimension is 10000.
2. The parameters used for the export is passed from user input

#### 💡 Related Function

`helper.EE_Export_Manager._export_direct_download` : Generate a data download link for direct download option

`helper.EE_Export_Manager.export_image` : Function to initialize earth engine export with input parameters validation while supporting several export option

## Google Cloud Storage Export

*To be developed*

## Google Drive Export

*To be developed*

## Export Task Monitoring

### Luma User Journey

1. The user can monitor the status of their export directly in Luma
2. If there's no update within 1 minute, they can refresh the task monitor to acquired the latest update of current task status
3. The user receive feedback regarding the status of the export task (in progress, complete, failed, etc)

### Luma Geospatial Engine

#### Related Function

`helper.EE_Export_Manager.get_task_status()`: Return task status using cache to reduce API calls

`helper.EE_Export_Manager.get_active_task()`: Return activate task information, especially the “ready” and “running” state

`helper.EE_Export_Manager.remove_task()`: Remove task from local tracking without affecting the task in earth engine task monitor

`helper.EE_Export_Manager.remove_completed_task()`: Remove all completed task for future monitoring. The removed task include all of the terminal states (complete, failed, or canceled export task)

**Part II**

**Luma Geospatial Engine**

# **1 Acquisition of Near-Cloud-Free Satellite Imagery**

# Module Overview

The output of this module is near-cloud-free satellite imagery that corresponds to the user-defined area of interest, which is essential for generating accurate and reliable land use and land cover (LULC) datasets. This module emphasizes the generation of pre-processed and corrected satellite imagery, incorporating procedures such as cloud masking and shadow removal.

# Input

| Name of Input                                   | Input Type     | Details               |
|-------------------------------------------------|----------------|-----------------------|
| Area of Interest                                | User's input   |                       |
| Target map date                                 | User's input   | YYYY or<br>DD-MM-YYYY |
| Desired spatial resolution                      | User's input   | in meters             |
| Maximum allowable cloud cover                   | User's input   | in percentage         |
| Raw satellite imagery                           | System's input |                       |
| Area of Interest from administrative boundaries | System's input |                       |

# Output

1. Selected AOI.
2. Near-cloud-free satellite imagery.

# Process

## 1.1 Selection of AOI

### 1.1.1 AOI from Shapefile

This sub-step does not involve any operations within the Luma Geospatial Engine.

### 1.1.2 AOI from Indonesia Administrative Boundaries

In addition to user define upload the system supports the selection of AOI from Indonesia administrative boundaries at regency level.

1. Load `ee.FeatureCollection` containing Indonesian administrative boundaries from Badan Informasi Geospasial. The data should be uploaded as GEE asset and the link for it must be provided and shared publicly
2. The system extracts the regency name and provide complete list of the available regencies for the user
3. The system convert the `ee.FeatureCollection` to `geodataframe` for visualization, while using the unconverted one as an AOI for the imagery search

#### Related Function

```
input_utils.load_regency_asset: Fetching GEE asset containing Indonesia adminis-  
trative boundaries  
input_utils.get_regency_name: Creating a list of regency name for the user  
input_utils.get_regency_geometry: Get geometry for the selected regency  
input_utils.convert_gee_features_to_gdf: convert ee.FeatureCollection to  
geodataframe for visualization in the system
```

## 1.2 Correction of Shapefile Geometries

### 1.2.1 Correction of Shapefile Geometries

1. The system performs a multi-step geometry sanitization process before converting any shapefile or GeoDataFrame into Earth Engine objects. **These input validation processes for shapefiles are designed mainly for Streamlit compatibility.** Each correction is handled by exactly one dedicated function:

- The Coordinate Reference System (CRS) is automatically corrected and reprojected to WGS 1984 (EPSG:4326).

#### Related Functions

`input_utils.shapefile_validator.fix_crs()`: to resolve coordinate reference system issues.

- Invalid, empty, or topologically broken geometries are detected and repaired.

#### Related Functions

`input_utils.shapefile_validator.clean_geometries()`: to repair invalid geometries.

- An initial structural validation and basic geometry fixing.

#### Related Functions

`input_utils.shapefile_validator.validate_and_fix_geometry()`: to validate and fix geometries issues.

- Polygon-specific validation, including removal of invalid coordinates and simplification of overly complex shapes.

#### Related Functions

`input_utils.shapefile_validator.validate_polygons()`: to checks point geometries, remove invalid coordinates, and simplify polygon geometries.

- Every individual coordinate is checked to ensure it falls within valid global latitude/longitude ranges.

#### 💡 Related Functions

`input_utils.shapefile_validator.is_valid_coordinate()`: to checks point geometries, remove invalid coordinates, and simplify polygon geometries.

- The number of vertices in each geometry is counted and logged (for diagnostic and simplification decisions).

#### 💡 Related Functions

`input_utils.shapefile_validator.count_vertices()`: to checks point geometries, remove invalid coordinates, and simplify polygon geometries.

- Finally, a comprehensive pre-upload validation is executed to guarantee that every geometry is valid, non-empty, correctly typed, and fully compatible with Earth Engine.

#### 💡 Related Functions

`input_utils.shapefile_validator.final_validation()`: to checks point geometries, remove invalid coordinates, and simplify polygon geometries.

2. After the system succeeded to validate the shapefile, the system converted the GeoDataFrame file format into Earth Engine geometry format.

#### 💡 Related Functions

`input_utils.EE_converter.convert_aoi_gdf()`: to conduct the conversion of `.gdf` input file into `ee.geometry`. Three different conversion approaches was defined in this function:

- Primary method: using `geemap.gdf_to_ee()`
- If primary method fails, proceed to secondary method: manual GeoJSON conversion by union multiple geometries first (`gpd.GeoDataFrame([{'geometry': union_geom}], crs=gdf.crs)`) and convert it to GeoJSON (`gdf.to_json()`). **Only polygon and multipolygon geometry type is supported in this method.**
- If both primary and secondary method fails, proceed to the last option: bounding box conversion to estimate the rectangular approximation of the AOI (`ee.Geometry.Rectangle([bounds[0], bounds[1], bounds[2], bounds[3]])`).

## 1.3 Selection and Preparation of Satellite Imagery

### 1.3.1 Satellite Data Acquisition and Filtering

1. The system specifies Landsat data collections for optical processing.

#### 💡 Related Functions

`data_acquisition.Reflectance_Data.OPTICAL_DATASETS` defines the properties of Landsat Surface Reflectance (SR) collections (Collection 2, Level-2) to be fetched for optical data processing. In this list, the metadata for reporting the data retrieval is also define. More satellite sensor can be define here

`data_acquisition.Reflectance_Data.THERMAL_DATASETS` Define the properties of Landsat Top-of-Atmosphere (TOA) thermal bands. Since some landsat mission have different band names designation, the thermal band is refer in the list

The system currently supports Landsat 1-3 RAW collection and and Landsat 4-9 surface reflectance (SR) collection. Additionally, Landsat 4 - 9 Top-of-Atmosphere (TOA) is used for retrieving thermal band as additional input band for the classification. This consideration stems from Landsat SR collection thermal band quality which often resulted in no data and prone to over-correction in high contrast land cover feature. Landsat mission availability is as follows:

- Landsat 1 Multispectral Scanner/MSS (1972 - 1978)
  - Landsat 2 Multispectral Scanner/MSS (1978 - 1982)
  - Landsat 3 Multispectral Scanner/MSS (1978 - 1983)
  - Landsat 4 Thematic Mapper/TM (1982 - 1993)
  - Landsat 5 Thematic Mapper/TM (1984 - 2012)
  - Landsat 7 Enhanced Thematic Mapper Plus/ETM+ (1999 - 2021)
  - Landsat 8 Operational Land Imager/OLI (2013 - present)
  - Landsat 9 Operational Land Imager-2/OLI-2 (2021 - present)
2. The system retrieves images that match user's selected criteria, including spatial resolution, year, and cloud cover based on the following input parameters: AOI, target map date, and cloud cover threshold.

#### 💡 Related Functions

`data_acquisition.parse_date_input()`: Parse date input to YYYY-MM-DD format. Accepts either a year (int or 4-digit string) or a full date string. For year inputs, returns Jan 1 for start dates or Dec 31 for end dates. This function is used by `get_optical_data` and `get_thermal_bands`

`data_acquisition.Reflectance_Data.get_optical_data()`: to retrieve and the

Landsat image collection based on the input parameters. This function uses the `mask_landsat_sr`, `apply_scale_factors`, and `rename_landsat_bands` to return a masked, scaled, and renamed image collection.

- The system performs cloud masking using the quality assurance band (QA\_PIXEL) QA\_PIXEL band that encodes pixel conditions as bitwise flags using specific bit flags. Deterministic bit tests are applied to identify and exclude pixels affected by cirrus, clouds and cloud shadows. The bit value is adjusted based on the Landsat sensor selected by the user. Landsat 8-9 QA\_PIXEL stored per-pixel confidence levels for clouds cirrus, and clouds shadow. These confidence values (ranging from 0–3) are extracted and filtered using hard-coded thresholds, such that pixels with medium to high confidence are masked out. Other Landsat sensor did not have confidence bit, therefore, only deterministic bit are used for cloud masking

 Related Functions

`data_acquisition.Reflactance_Data.mask_landsat_sr()`: Mask clouds, shadows and cirrus for Landsat Collection 2 SR using QA\_PIXEL band with confidence threshold

- The system applies scaling to the optical reflectance bands to convert the raw digital numbers (DNs) into physically meaningful surface reflectance values. A scale factor 0.0000275 and offset -0.2 are applied, as defined in the Landsat Collection 2 Surface Reflectance metadata. This conversion standardize the reflectance values to a range of approximately 0 to 1, enabling accurate comparison and analysis across different scenes and sensors.

 Related Functions

`data_acquisition.Reflactance_Data.apply_scale_factors()`: to apply Landsat collection 2 scaling factors using the following formula: Digital Number (DN) \* scale\_factor + offset. The scale factor will only be applied to optical bands.

- The system renames the bands accordingly with the sensor type input.

 Related Functions

`data_acquisition.Reflactance_Data.rename_landsat_bands()`: to standardize Landsat Surface Reflectance (SR) band names based on sensor type. From 'SR\_B' or 'B' to 'NIR', 'GREEN', etc.

3. The system then use the same parameter as multispectral data and use it to search

collection 2 TOA data, except for Landsat 1-3 MSS. The TOA data thermal band (namely, band 10) is then stacked with multispectral band from SR data. Since band 11, contain larger calibration uncertainty, only band 10 is used. No additional pre-processing is conducted for thermal band since digital number value alone is considered satisfactory for discriminating land cover features.

#### Related Functions

`data_acquisition.Reflectance_Data.has_thermal_capability()`: Function to check if the landsat mission selected contain thermal bands  
`data_acquisition.Reflectance_Data.get_thermal_bands()`: to retrieve thermal band from landsat collection 2 TOA data

4. The system evaluates whether the retrieved imagery meets the required spatial resolution, input year, and cloud cover thresholds. If suitable images are found, the system generates a composite by calculating the median value across the selected imagery, as implemented using the `median()` function from Earth Engine. The system also then clips and mosaics the satellite imagery.

## 1.4 Mosaic and Composite for Satellite Imagery

The final image (single layer) which will be presented to the user is created using `final_image` methods, which resulted in mosaic imagery or temporal aggregation composite imagery.

### 1. Mosaicking

Mosaicking approach simply stacks all images in the collection and, for each pixel, selects the value from the topmost image according to the collection's ordering. The system implement quality mosaic, which allows the use of quality band that affected the order of mosaic

#### Related Functions

`data_acquisition.final_Image.get_quality_mosaic` create a quality mosaic image based on quality band (source band). `add_quality_band` compute or select the band that will be used for creating the mosaic. Three quality band is supported, namely NDVI and NIR. The mosaic is created based on the highest pixel value of the corresponding band

### 2. Temporal Aggregation

Temporal aggregation used statistical reducers, (such as median) to combine pixel values from an image collection over time. A **median composite** often produces a noticeably

cleaner image because cloud and other high-reflectance features tend to occupy the upper tail of the distribution. This approach, naturally excludes the feature, resulting in fewer artifacts and a more consistent surface reflectance. However, if at certain location there's no valid pixel across the image collection, the location will result in no-data (blank region). Temporal aggregation function is paired with `calculate_coverage` to provide report of the valid pixels in an AOI.

#### 💡 Related Functions

`data_acquisition.final_Image.get_temporal_composite`: Create a temporal aggregation composite using earth engine reducers. Supported reducers are `median`, `mean`, `min`, `max`, and `percentile`

### 3. Valid Pixel Reporting

To provide comprehensive report on the final image creation, system provide the data range which composite is created. Additionally, valid pixel and no data is estimated and reported to the user.

#### 💡 Related Functions

`calculate_coverage` estimate, the valid pixels (unmasked by cloud masking) within AOI. This quantified the 'no data' or blank blocks on the final image. This served as alternative approach in reporting the cloud cover within AOI, since in theory, the cloud cover within AOI is already masked out during cloud masking procedure. This function is used within the `get_temporal_composite` and reported in the final image creation.

The `calculate_coverage` serves as an alternative to calculate cloud cover within AOI. Since cloud cover should be masked out during cloud masking and remove during temporal aggregation

## 1.5 Visualization and Saving of Processed Satellite Imagery

### 1.5.1 Visualization and Saving Processed Satellite Imagery

1. The system collects the statistics of the satellite images found.

#### 💡 Related Functions

`data_acquisition.Reflectance_Stats.get_collection_statistics()`: to get comprehensive statistics about the gathered image collection.

2. The system provide a preset of commonly used band combinations as well as optimal value range for the imagery. The commonly used composites define in Luma-GE are:

1. True Color Composites (RED, GREEN, BLUE)
2. False Color Infrared Composites (NIR, RED, GREEN)
3. Short-wave Infrared Composites (SWIR2, NIR, RED)
4. Land/Water (NIR, SWIR1, RED)

 Related Function

`data_acquisition.Vis_Params.BAND_COMBINATIONS`: The list of preset band combinations as well as their default value

`data_acquisition.Vis_Params.get_combinations_names()`: function to fetch band combinations names from the preset

`data_acquisition.Vis_Params.get_vis_params()`: function to retrieve band combinations and visualization parameters from the define list

3. The system provide tools for the user to build their own band combinations and modified the imagery minimum/maximum value as well as the gamma value, improving the visualization of the data

 Related Function

`data_acquisition.build_custom_vis_param()`: function to allow the user to build their own band combinations

4. For details on exporting to Google Cloud Storage, refer to Earth Engine Initialization and Authentication page.
5. The system saved the satellite imagery as an object to be loaded in the next modules.

## **2 LULC Classification Scheme**

## Module Overview

This module prompts the user to define the list of land cover or land use class which they're going to classify using the Luma platform. Luma supports user-defined classification schemes and default classification or templates for commonly used classification schemes.

# Input

| Name of input                       | Input Type     | Details                                                           |
|-------------------------------------|----------------|-------------------------------------------------------------------|
| User's LULC list of classes         | User's input   | Entered with <code>.csv</code> file or manually defined on the UI |
| Other existing LULC list of classes | System's input | RESTORE+ LULC class                                               |

# Output

LULC list of classes to be applied in the final classification result

# Process

## ! Error Handling Notification

This module cannot be accessed if the system does not detect any saved satellite imagery from [Module 1](#).

## 2.1 Selection of Classification Scheme

This sub-step does not involve any operations within the Luma Geospatial Engine.

- If the user chooses to upload their own schema, the system proceeds to [Section 2.1.1](#)
- If the user chooses to define the classification manually on the UI, the system proceeds to [Section 2.1.2](#)
- If the user chooses a default classification scheme, the system proceeds to [Section 2.1.3](#)

### 2.1.1 User Uploading Their Own Classification Scheme

1. The system automatically detects column headers based on common naming conventions for ID, class name, and color code. If the expected headers are not found, the user must manually assign the correct column names.

#### 💡 Related Functions

`classification_scheme.LULC_scheme_Manager.auto_detect_csv_columns()`: this function automatically detects the column headers. The set for the common naming conventions are currently hardcoded as the following:

- For ID: “id”, “classid”, “kode”, “code”
- For class name: “classname”, “class”, “kelas”, “name”, “nama”
- For color code: “color”, “colorcode”, “warna”, “colour”, “hex”, “palette”

2. The system validates the uploaded `.csv` to ensure IDs are numeric and unique, applies a set of distinct colors if no color code was provided by the user. The user can change the color code manually.

#### 💡 Related Functions

`classification_scheme.LULC_scheme_Manager.process_csv_upload()`: to validate class data by checking ID integrity, assigning colors (`_generate_distinct_colors`), and returning a success status with a summary message.

`classification_scheme.LULC_scheme_Manager._generate_distinct_colors()`: to generate a predefined set of distinct colors for LULC classes (up to 20). If the number of classes exceeds this limit, the method falls back to generating random colors via `_generate_random_colors()`.

`classification_scheme.LULC_scheme_Manager._generate_random_colors()`: to generate random hex color code.

`classification_scheme.LULC_scheme_Manager.finalize_csv_upload()`: to finalize the CSV file by applying any user-assigned colors, saving and sorting the class list, and clearing temporary data.

### 2.1.2 User Entering the Classification Manually

1. The system validates user-provided class input for numeric ID, non-empty class name, and uniqueness (if adding new classes).

#### 💡 Related Functions

`classification_scheme.LULC_scheme_Manager.validate_class_input()`: Validates class ID and name, checks for uniqueness, and returns a validity flag with an error message if invalid.

`classification_scheme.LULC_scheme_Manager.add_class()`: Adds a new class or updates an existing class, validates inputs, assigns colors, sorts classes, and updates the next available ID.

2. The system normalizes class name inputs through a sequential process: it first applies Unicode normalization using the NFKD form, converts all characters to lowercase, replaces common separators (`_`, `-`) with spaces, removes non-alphanumeric characters, collapses multiple consecutive spaces, and applies light lexical normalization to reduce superficial grammatical differences, such as singular versus plural forms (e.g. “forest” and “forests”). Missing values (`None` or `NA`) are returned as `None`. Optionally, words within the class name can be sorted alphabetically to enable order-independent matching (e.g. “Forest Mixed” = “Mixed Forest”).
3. Temporary edit mode flags are managed to differentiate between adding and updating classes.

#### Related Functions

```
classification_scheme.LULC_scheme_Manager.edit_class()  
classification_scheme.LULC_scheme_Manager.delete_class()  
classification_scheme.LULC_scheme_Manager.cancel_edit()
```

### 2.1.3 User Selecting a Default Classification Scheme

1. The system loads the chosen classification scheme classes.

#### Related Functions

```
classification_scheme.LULC_scheme_Manager.load_default_scheme():    to  
load an existing classification scheme from a dataset.  
classification_scheme.LULC_scheme_Manager.get_default_schemes():  
stores the RESTORE+ LULC class in a dictionary.
```

2. If the user selects a subset of the default classes, the system stores the selection into a variable that will be used for the reclassification process in [Module 6](#).

#### Related Functions

```
classification_scheme.LULC_scheme_Manager.store_classes_of_interest():  
to store the user's class of interest as a variable that will be used for the reclassifica-  
tion process in Module 6.
```

## 2.2 Saving and Downloading the LULC Classification Table

### 2.2.1 Saving and Downloading the LULC Classification Table

1. The system generates a .csv file containing the final LULC classification scheme table.

#### Related Functions

```
classification_scheme.LULC_scheme_Manager.get_csv_data() : to generate a  
.csv file containing the final classification table that has been standardized by  
get_dataframe.  
classification_scheme.LULC_scheme_Manager.get_dataframe() : to standard-  
ize the column names of the LULC classification table to "ID", "Land Cover Class",  
"Color Palette"
```

2. The system checks whether the user has defined at least one class.

#### Related Functions

`classification_scheme.LULC_scheme_Manager.has_classes()`: to check if any classes are defined.

`classification_scheme.LULC_scheme_Manager.get_class_count()`: get the information of the number of the defined classes.

3. The system saved the LULC class classification scheme as an object to be loaded in the next modules.

## **3 Sample Data Generation**

# Module Overview

To be developed.

# Input

| Name of input                     | Input Type               | Details              |
|-----------------------------------|--------------------------|----------------------|
| User's sample data                | User's Input             |                      |
| Default sample data               | System's Input           | RESTORE+ sample data |
| LULC classification class table   | Input from Other Modules | Output from Module 2 |
| AOI                               | Input from Other Modules | Output from Module 1 |
| Near-cloud free satellite imagery | Input from Other Modules | Output from Module 1 |

# Output

Statistical analyses of the sample data.

# Process

## 3.1 Checking Prerequisites from Previous Modules

### 3.1.1 Checking Prerequisites from Previous Modules

This sub-step does not involve any operations within the Luma Geospatial Engine.

## 3.2 Sample Data Generation

### 3.2.1 User Uploads Their Own Sample Data

1. The system verifies the format data input of the `.shp` inside the `.zip`.
2. Once the input format data is validated, the system runs Section [3.2.2](#)
3. The system loads the sample data after verifying the upload
  - Check the initial sample count and conduct sample filtering based on its geometry, making sure all of the sample is located inside the area of interest
  - Handle a large dataset by stratified sampling so that the sample does not exceed 5000 features, and maintain class distribution representation
  - Convert the shapefile into a geodataframe
  - Updates the class field identifier in the training data dictionary.

#### Related Function

`sample_data.SyncTrainData.LoadTrainData()`: to load the sample data into the system by conducting several checks

#### Feature note

The main issue is that the original stratified sampling algorithm made sequential `.getInfo()` calls for each class label when calculating class sizes and extracting features, which could cause browser crashes or exceed Earth Engine quotas when pro-

cessing large datasets (5000+ features). The improved implementation consolidates all server-side computations into a single aggregated request, keeping all operations on the server side until the final `.getInfo()` call, thus avoiding excessive client-server communication and quota limitations.

### 3.2.2 Verifying Sample Data from User's Input

1. The system checks which column in the sample dataset should be used as the class label.

#### Related Functions

`sample_data.SyncTrainData.SetClassField()`: Set the class name field for sample data.

2. The system validates class labels in the sample dataset by filtering out class values that do not exist in the classification scheme from **Module 2**, and updates the sample data and validation report accordingly.

#### Related Functions

`sample_data.SyncTrainData.ValidClass()`: Validate classes in sample data.

3. The system checks the number of sample data for each class. The minimum required sample size per class is currently hard-coded at 20.

#### Related Functions

`sample_data.SyncTrainData.CheckSufficiency()`: Check if there are sufficient samples per class.

4. The system filters the training samples so only those that spatially overlap the AOI remain, updates validation counts, and records warnings if AOI filtering fails.

#### Related Functions

`sample_data.SyncTrainData.FilterTrainAoi()`: Check if there are sufficient samples per class.

5. The system generates a summary table of the user's uploaded sample data. The summary includes:

- LULC class ID

- LULC class name
- Number of sample data of the LULC class
- Percentage of the sample data count
- Sufficiency category of the LULC class' sample data

#### 💡 Related Functions

`sample_data.SyncTrainData.TrainDataRaw()`: Check if there are sufficient sample data per class.

### 3.2.3 On-Screen Sampling

This sub-step does not involve any operations within the Luma Geospatial Engine. The on-screen sampling feature was developed using `folium` for the Streamlit development.

### 3.2.4 Using Default Sample Data

1. The system retrieves an existing sample data accordingly with the chosen LULC scheme. In this development phase, the RESTORE+ sample data is retrieved from Google Earth Engine asset.

## 3.3 Sample Data Preview

1. The system runs Section [3.2.2](#) for the **updated** sample data.
2. The system saved the sample data as an object to be loaded in the next modules.

## **4 Sample Data Quality Analysis**

## Module Overview

This module assesses the spectral separability of LULC classes using sample data and near-cloud-free satellite imagery. Pairwise class separability is quantified using the Transformed Divergence (TD) method based on pixel-level spectral statistics, using only the spectral data from Module 1.

This module is diagnostic only and does not produce inputs for subsequent modules. Its outputs support interpretation of class confusion and provide justification for feature enhancement in later stages of the workflow.

# Input

| Name of Input                            | Input Type               | Details                                                                                                                                                                                                                                              |
|------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameters for the separability analysis | User's input             | Contains: <ul style="list-style-type: none"> <li>• Spatial resolution of the imagery from Module 1 can be adjusted for the separability analysis</li> <li>• Number of maximum pixels which the analysis will run for each pair of classes</li> </ul> |
| Sample dataset                           | Input from Other Modules | Output from Module 3                                                                                                                                                                                                                                 |
| Near-cloud free satellite imagery        | Input from Other Modules | Output from Module 1                                                                                                                                                                                                                                 |

# Output

1. Separability analysis result.
2. Spectral profile visualization and text description.

# Process

## ! Error Handling Notification

This module cannot be accessed if the system is missing the required inputs from the Input from Other Modules.

## 4.1 Specifying the Separability Analysis Parameter

This sub-step does not involve any operations within the Luma Geospatial Engine.

## 4.2 Conducting the Separability Analysis

### 4.2.1 Initialize the Separability Analysis

This sub-step does not involve any operations within the Luma Geospatial Engine.

### 4.2.2 Validating the sample data's geometry

1. The system validates the geometry of the sample data stored in the system from Module 3 output.

#### 💡 Related Function

`input_utils.shapefile_validator.validate_and_fix_geometry()`: to validate and fix geometries issues.

## ! Error Handling Notification

An error message is displayed if the system failed to run `validate_and_fix_geometry()`.

### 4.2.3 Converting sample data into Earth Engine format

1. The system converts sample data from `.gdf` data into Earth Engine Feature Collection format

#### Related Function

`input_utils.shapefile_validator.convert_roi_gdf()`: to convert geo-dataframe into EE Feature Collection. The supported multi-geometries type for this conversions are `MultiPoint` and `MultiPolygon`.

#### Error Handling Notification

An error message is displayed if the system failed to run `convert_roi_gdf()`.

### 4.2.4 Starting the separability analysis computation

1. The system initializes the Related Function's for separability analysis with the appropriate required input objects for the `class`.

#### Related Function

`sample_data_quality.sample_quality()`: a `class` to handle all the workflows for separability analysis.

### 4.2.5 Counting sample data statistics

1. The system retrieves the basic statistic information of the sample data.

#### Related Function

`sample_data_quality.sample_quality.sample_stats()` : to retrieve the sample data's total sample count, unique classes, class-wise sample counts, and class balance, and includes class names when provided.

`sample_data_quality.sample_quality.get_sample_stats_df()` : to turn the statistic information into a dataframe format.

### 4.2.6 Extracting spectral value

1. The system extracts the pixel-level spectral values for each sample data by sampling the composite image at the specified scale. The system also limits the maximum number of pixels per LULC class that will be extracted to avoid memory overload.

#### Related Function

`sample_data_quality.sample_quality.extract_spectral_values()` : to extract and compile spectral reflectance values for all sample data. The result is a dataframe containing band values for each sample data.

`sample_data_quality.sample_quality.limit_samples_per_class()` : limits (down-samples) the number of sample data per class whenever the number of available samples exceeds the maximum pixel threshold with random sampling.

### 4.2.7 Calculating the extracted spectral feature statistics

1. The system computes detailed pixel-level statistics for each LULC class (mean, standard deviation, minimum, maximum, median, and sample count) based on the extracted spectral values and formats these results into a structured summary table.

#### Related Function

`sample_data_quality.sample_quality.sample_pixel_stats()` : calculates per-class spectral statistics (mean, std, min, max, median, count) for all bands.

`sample_data_quality.sample_quality.get_sample_pixel_stats_df()` : converts the computed statistics into a dataframe.

### 4.2.8 Calculating separability index

1. The system computes the class-to-class separability using Transformed Divergence (TD).
2. The system generates a pairwise separability matrix and converts the results into a structured dataframe.
3. The system categorizes the number of good, weak, and poor class pairs as a summary for user.

#### 💡 Related Function

`sample_data_quality.sample_quality.check_class_separability()`: calculates the pairwise separability values between classes using Transformed Divergence. The choice of using TD as the algorithm is still hard-coded.

`sample_data_quality.sample_quality.get_separability_df()`: converts separability results into a dataframe with class names and interpretation levels.

`sample_data_quality.sample_quality.lowest_separability()`: extracts the class pairs with the lowest separability scores.

`sample_data_quality.sample_quality.sum_separability()`: produces a summary report counting how many class pairs fall under good, weak, or poor separability categories.

## 4.3 Visualization of Reference Data's Spectral Profile

1. The system provides four different visualization options of the spectral distributions and clustering.

#### 💡 Related Functions

`sample_data_quality.spectral_plotter.plot_histogram()`: generates overlaid interactive histograms of selected bands showing class-wise frequency distributions

`sample_data_quality.spectral_plotter.plot_boxplot()`: creates interactive boxplots summarizing per-class band statistics (median, quartiles, outliers) for selected bands.

`sample_data_quality.spectral_plotter.interactive_scatter_plot()`: produces an interactive 2D scatter plot of two bands with class coloring and hover details for exploring class clustering.

`sample_data_quality.spectral_plotter.static_scatter_plot()`: renders a static 2D scatter plot with optional confidence ellipses

`sample_data_quality.spectral_plotter.add_elipse()`: draws a confidence ellipse axis for a class' 2D band distribution (used by `static_scatter_plot`).

`sample_data_quality.spectral_plotter.scatter_plot_3d()`: builds an interactive 3D scatter plot to explore three-band feature space and rotate/zoom clusters.

## 5 Predictor Generation

## Module Overview

The aim of Module 5 is to derive a set of predictors that optimize the discrimination of LULC classes in the classification model. Predictors are generated from three principal categories: spectral transformations (indices), terrain-based metrics, and distance-based metrics, capturing complementary spectral, topographic, and spatial information. These features are critical because raw spectral data alone often fail to adequately separate spectral or spatially similar classes, and their inclusion enhances model accuracy and robustness.

The module outputs a comprehensive predictor stack that serves as input for LULC map generation in Module 6.

# Input

| Name of Input                      | Input Type               | Details                                                |
|------------------------------------|--------------------------|--------------------------------------------------------|
| supported spectral transformations | System's input           | Retrieve the pre-define function from existing library |
| Terrain-based metrics              | System's input           |                                                        |
| Distance-based metrics             | System's input           |                                                        |
| AOI                                | Input from Other Modules | Output from Chapter <a href="#">1</a>                  |
| Near-cloud free satellite imagery  | Input from Other Modules | Output from Chapter <a href="#">1</a>                  |
| Sample data                        | Input from Other Modules | Output from Chapter <a href="#">3</a>                  |

# Output

Covariate stack used in the classification model (Module 6)

# Process

## 5.1 Selection of Predictors

1. The system presents a list of available predictor categories, which include:
  - Spectral transformation (indices)
  - Terrain-based metrics
  - Distance-based metrics

## 5.2 Calculation of Predictors

1. The system retrieves the predictors based on user's selection. The source of the predictors are as the following:
  - Terrain-based metrics
    - **Elevation:** The system fetch Digital Elevation Model (DEM) from GEE data catalog. Several DEM is supported by Luma-GE, namely [NASADEM](#), [Copernicus GLO-30](#), and [ALOS DSM V4](#). The default choice for the DEM is NASADEM since it provide higher vertical accuracy compared to other DEM product. All of the aforementioned DEM product have the spatial resolution of 30 m. Matching the spatial resolution of Landsat data primarily use in Luma
    - **Slope:** Calculated from selected DEM using `ee.Terrain.slope` function. Slope unit is presented in degree units (0-90)
    - **Aspect:** Similar to slope, this data is calculated from the selected DEM, using `ee.Terrain.aspect`. The units are degree where 0=N, 90=E, 180=S, 270 =W

### Related Functions

`predictor.terrain_calculator.calculate_elevation` : Core function to fetch DEM data from GEE catalog.  
`predictor.terrain_calculator.calculate_slope` : Core function to calculate slope from selected DEM data

```
predictor.terrain_calculator.calculate_aspect : Core function to calculate aspect data from selected DEM data
```

- Distance-based metrics

The system calculates distance from several features that might affect the presence of certain land cover class. The selection and distance calculation of these features are inspired by RESTORE+ project. Distances can be measured from various infrastructures, primarily roads, coastlines, and settlements.

 Related Functions

```
predictor.distance_calculator.calculate_distance_metrics: Core function to initialize the distance calculation logic for coastline, road, and settlement
```

Currently, the system support distance calculation from the following source:

- Coastline: [Natural Earth 10m coastline data](#),
- Settlement: [Facebook High Resolution Settlement Layer \(HRSL\)](#)
- Road: Indonesian Topographic Map (*Rupa Bumi Indonesia*) and OpenStreetMap data

- Spectral transformation indices

The system retrieves various spectral indices from the [Awesome Spectral Indices \(ASI\)](#), a standardized, ready-to-use curated list of spectral indices that can be used as expressions for computing spectral indices in remote sensing applications. Further reading regarding this library refers to [Montero \(2023\)](#). The following indices have been adapted for Luma-GE:

- Vegetation-related indices:
  - \* NDVI (Normalized Difference Vegetation Index)
  - \* EVI (Enhanced Vegetation Index)
  - \* SAVI (Soil Adjusted Vegetation Index)
  - \* MSAVI (Modified Soil Adjusted Vegetation Index)
  - \* OSAVI (Optimized Soil Adjusted Vegetation Index)
  - \* ARVI (Atmospherically Resistant Vegetation Index)
  - \* GBNDVI (Green-Blue Normalized Difference Vegetation Index)
  - \* GNDVI (Green Normalized Difference Vegetation Index)

- Water-related indices:
  - \* MNDWI (Modified Normalized Difference Water Index)
  - \* NDMI (Normalized Difference Moisture Index)
  - \* AWEInsh (Automatic Water Extraction Index with Shadow Elimination)
- Soil / built-up-related indices:
  - \* NDBI (Normalized Difference Build-Up Index)
  - \* DBSI (Dry Bareness Soil Index)
  - \* MBI (Modified Build-up Index)

#### 💡 Related Functions

`predictor.SpectralCalculator.get_avaliabile_indices`: Return the list of supported indices. This list contain information regarding the Index name, long-name, formula, and references

`predictor.SpectralCalculator.get_index_coefficients`: Get default coefficient values for a spectral index that needs user-define coefficients and parameters

`predictor.SpectralCalculator.validate_coefficients`: Conducts checks the provided coefficients are within reasonable ranges and provides warnings for unusual values that might indicate errors.

`predictor.SpectralCalculator.validate_index_requirements` : Validate that required bands are available for index computation.

`predictor.SpectralCalculator.calculate_indices_with_collection`: Conducts calculation for each scene (image collection) and then aggregate the result using mean reducers. This logic is used to preserve phenological pattern of vegetation and other land features which ultimately loss during if the final image composite is used.

`predictor.SpectralCalculator.get_calculation_summary` : provide summary of the calculation process of predictors :::

2. The main orchestration of predictor calculation is conducted after the system validate the required input. The system combines the additional predictors chosen from this module with the multispectral bands from Module 1.

#### 💡 Related Functions

`predictor.PredictorCalculation.validate_prerequisites`: Validate that required inputs are available for predictor computation. Required input are composite (`ee.Image`), AOI (`ee.FeatureCollection`) , and raw image collection

```
(ee.Image.Collection)
predictor.PredictorCalculation.compute_predictors : Initialize predictor
computation by validating inputs, calculating selected terrain and spectral metrics,
and stacking them into a final output.
predictor.PredictorCalculation.prepare_module6_summary: Prepare sum-
mary metadata for module 6
```

## 5.3 Predictors Visualization and Analysis

This section allow the user to visualize the resulted predictors to visually inspect the spatial distribution of the data.

1. If the option to conduct correlation analysis was selected by the user, the system conduct correlation analysis using the following approach
  - Creates a random sampling across the AOI (default value is 5000 points)
  - Extract the pixel value of the computed predictors. The spectral band predictors are not extracted, since the analysis is design to analyze the correlation between ancillary predictors
  - Computes correlation analysis using GEE's core function
2. The correlation analysis is conducted across all predictors, including the original multi-spectral band

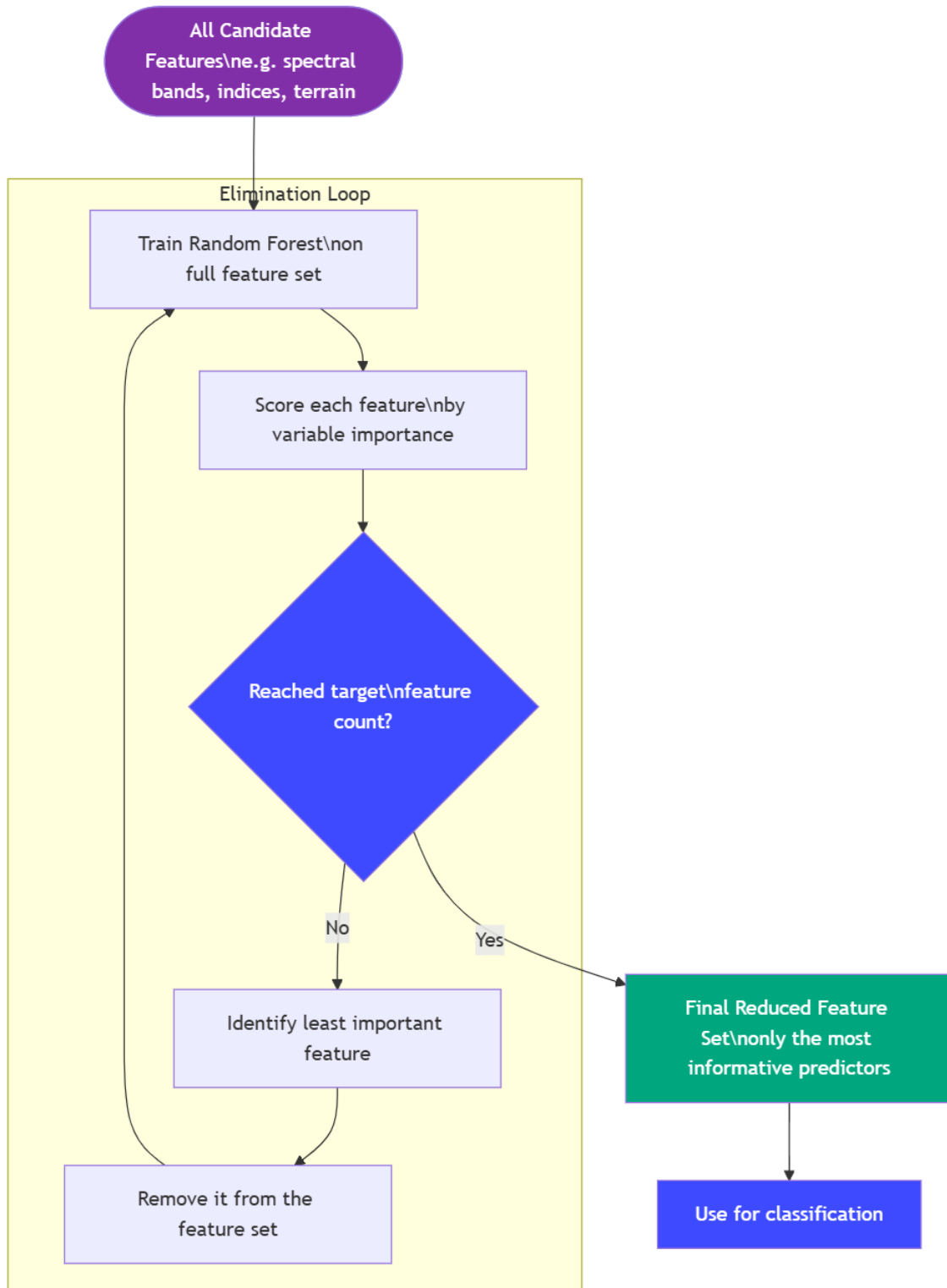
### Related Functions

```
predictor.EECorrelationAnalysis.generate_random_sampling: Generate ran-
dom sample across the AOI. Use default value of 5000 sample
predictor.EECorrelationAnalysis.compute_correlation: Main function to
compute Pearson correlation analysis based on the random samples generated.
Return the panda data frame use for visualization.
predictor.EECorrelationAnalysis.visualize: Visualize the correlation analy-
sis result using heatmap
```

## 5.4 Semi Automatic Predictor Selection Optimization

Evaluating the user's predictor selection is crucial step in determining the best combination of predictors that resulted in the optimal model performance. This approach is more commonly known as feature engineering in machine learning terminology. One of the commonly use

feature engineering approach is Recursive Feature Elimination (RFE). RFE applied a backward step-wise selection process to find optimal combination of predictors based on its contribution to the model. The RFE starts with training the model with all input predictors and discards the weakest predictor one iteration at a time, while re-evaluating their importance after each removal. The workflow for RFE is shown in the figure below:



1. The system perform input validation prior to RFE

- If the input is invalid, provide informative error messages. The invalid input includes missing data (training or testing) or missing ancillary predictors
- If the input is valid, proceed with RFE calculation

 Related Function

`predictor.RfeGee.validate_inputs()`: validate all input require for RFE calculations, namely: training data (`FeatureCollection`), test data (`FeatureCollection`), number of feature selected ( $>1$ ), and predictors (`ee.Image`)

2. The system perform RFE with default configuration and report the optimal predictors to the user.

- the default configuration are as follow:
  - Minimum feature to select: 10
  - Minimum random forest tree: 100
  - Training data (for RFE): 70% of user input
  - Test data (for RFE): 30% of user input
- Since Earth Engine did not have default feature selection algorithm, the system create looping mechanism that train random forest classifier and return feature importance ranking and accuracy of the model for each iterations. The code is inspired by similar approach by [this github repository](#)

 Related Function

`predictor.RfeGee._train_classifier()` : core function to perform random forest classification with default configuration.

`predictor.RfeGee._get_importance_and_accuracy()` :Retrieve feature importance test accuracy in a single round-trip to the Earth Engine servers. Optimizing the time required for finish the entire RFE.

`predictor.RfeGee._adaptive_step()` : fallback mechanism to reduce API calls in case computational loads become larger.

`predictor.RfeGee.fit()`: function to conduct the complete workflow for the entire RFE, including train classifier and calling importance and accuracy.

- To optimize API calls in Earth Engine servers, the calculations is conducted using server-side computations, and only called after all iterations is complete.

- The result of RFE is visualize using graphic and tables, including the selected features, rank of eliminations, and accuracy curve

#### 💡 Related Function

`predictor.RfeGee.get_selected_features` Function to retrieve optimal predictors list based on RFE result

`predictor.RfeGee.get_ranking` Function to rank importance of each predictors

`predictor.RfeGee.plot_accuracy_curve` Plot the accuracy curve as each features is remove from original input features

`predictor.RfeGee.plot_feature_rankings` Plot the feature/predictor importance from RFE calculations

`predictor.RfeGee.plot_rfe_summary` Create summary plot of accuracy curve and feature rankings

`predictor.RfeGee.get_results_dataframe` Generate a dataframe of the corresponding accuracy and importance data

- Currently, the user require to determine the number of selected predictors for RFE. Future development will optimize this features

## **6 LULC Map Generation**

## Module Overview

This module performs supervised Random Forest classification to generate a land cover map. It includes input review, hyperparameter definition, feature extraction, model training, hard/soft classification, feature importance, and accuracy evaluation using test data.

# Input

| Name of input                    | Input type               | Details                                                                                                          |
|----------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------|
| Sample data split proportion     | User's input             | Optional, if not set then use default value                                                                      |
| Random Forest (RF) parameters    | User's input             | Optional, if not set then use default value                                                                      |
| Satellite imagery within the AOI | Input from other modules | Module 1                                                                                                         |
| LULC classes                     | Input from other modules | Module 2, including the selection of the classes of interest if the user uses the default classification scheme. |
| Sample data                      | Input from other modules | Module 3                                                                                                         |

# Output

1. Trained LULC classification model.
2. Model accuracy report if validation data available (including overall accuracy, precision, recall, F1-score, Kappa, model error matrix). This output is only available if the user has a validation data from the **splitted the sample data**
3. Classified LULC raster map.

# Process

## ! Error Handling Notification

This module cannot be accessed if the system is missing the required inputs from the Input from Other Modules

## 6.1 Checking Prerequisites from Previous Modules

1. The system validates availability of the required inputs.

## 6.2 Extracting spectral feature from sample data

### 6.2.1 Splitting the sample data

1. The system conducts feature extraction for both training and validation data accordingly to the proportion set by the user.
2. The system uses stratified random split method to conduct the data splitting. This method randomly selects samples within each class to proportionally create training and testing sets while preserving class distribution.

## 💡 Related Functions

`classification.FeatureExtraction.stratified_split()` : performs stratified random split of sample data based on LULC class labels.

## 6.3 Creating classification model

### 6.3.1 Creating classification model

1. The system create the classification model using hard classification method.

#### Related Functions

`classification.Generate_LULC.hard_classification()` : trains a RF classifier (`smileRandomForest`) using extracted training data and applies it to the input imagery to produce a hard (pixel-level) multiclass.

### 6.3.2 Reviewing classification model's quality

1. The system extracts feature importance from the trained Random Forest model and presents a ranked table of bands (or a fallback equal-weight estimate if true importance is unavailable).

#### Related Functions

`classification.Generate_LULC.get_feature_importance()` : extracts and returns model feature importance as a dataframe; if the model explanation is not available, falls back to an equal-weight estimate.

2. The system calculates model evaluation metrics on the validation dataset. This option only available if the user perform data split.

#### Related Functions

`classification.Generate_LULC.evaluate_model()` :classifies the validation data with the trained classifier, builds the confusion matrix, and returns a dictionary of accuracy metrics

3. Confusion matrix and model accuracy summary are then rendered in the user-interface.

### 6.3.3 Visualizing LULC classification map result

1. The system loads the user's class of interest stored from Module 2.
2. The system reclassifies the default classification scheme's map into a map that only shows the user's class of interest and an additional class labeled as "Others".

#### Related Functions

`classification.Generate_LULC.reclassify_map_by_classes()`: reclassifies the default classification scheme final map to only show the user's class of interest and show the other classes as "Others".

#### **6.3.4 Downloading the LULC classification map result**

1. This sub-step does not involve any operations from Luma Geospatial Engine. Direct download settings are defined from Streamlit, using the same workflow as defined in [Section 1.5](#).
2. For details on exporting to Google Cloud Storage, refer to [Earth Engine Initialization and Authentication](#) page.

## **7 Thematic Accuracy Assessment**

## Module Overview

This module performs a comprehensive thematic accuracy assessment of the land cover map using an independent ground reference data. The framework for this module are based on the publication, *Good Practices for Estimating Area and Assessing Accuracy of Land Change*, by [Olofsson et al 2014](#). This framework separates accuracy assessment into three major components, namely sampling design, response design, and analysis. The Luma-GE only adapt the sampling design and analysis components. Since the response design consist of protocols and guideline in obtaining the ground reference data, this component are currently not implemented.

# Input

| Name of input       | Input type               | Details              |
|---------------------|--------------------------|----------------------|
| Validation map      | User's input             | In shapefile format. |
| Classified LULC map | Input from Other Modules | Module 6             |

# Output

- Confusion matrix.
- Thematic accuracy metrics: Overall Accuracy (OA), Kappa Coefficient, Producer's Accuracy (PA), User's Accuracy (UA).
- Reference data sites (optional, for users who do not have reference data, the system will generate reference data sites for them to label outside Luma-GE).
- Spatial distribution of error

# Process

## 7.1 Checking Prerequisites from Previous Modules

The system validates availability of the required inputs.

## 7.2 Thematic Accuracy Assessment

### 7.2.1 Selecting The Workflow

Two workflow are available for Module 7. The first workflow is design for the user who did not have reference data. The second workflow is for the user who already have a reference data.

For each workflow, the geospatial engine perform the following operation

### 7.2.2 Generate Reference Sample Workflow

This workflow consist of two steps, steps, namely sample size calculation and sample allocation. The result of each steps is how many minimum sample required for each class and the location for the samples.

#### Sample Size Calculation

Sample size calculation for stratified random sample is calculated using the formula provided by [Cochran \(1977, Eq. \(5.25\)\)](#). The samples are proportionally allocated for each strata (class), resulting in balance sample allocation. The key steps for sample size calculation is as follows:

1. Calculate stratum standard deviation:  $S_i = \sqrt{U_i \times (1 - U_i)}$

Where:

$S_i$  = standard deviation for stratum i

$U_i$  = expected accuracy for class

2. Calculate stratum weight:  $W_i = N_i / N$   
 $W_i$  = weight of stratum i  
 $N_i$  = Number of pixels in class i  
 $N$  = Total number of pixels
3. Calculate total sample size:  $n = (\sum(W_i \times S_i) / SE)^2$   
 $n$  = Total required samples  
 $SE$  = desire standard error
4. Allocate sample per stratum  $n_i = n \times (W_i \times S_i) / \sum(W_i \times S_i)$   
 $n_i$  = Number of samples for stratum i

#### Related Function

`sample_size_calculator.get_pixel_count_per_class`: calculate pixel count for each class using earth engine's reducer  
`sample_size_calculator.validate_sample_size_inputs`: Input validation prior to main sample size calculation  
`sample_size_calculator.calculate_strata_sample`: Main function that calculate sample size for each class

### Sample Allocation

The required sample for each class is allocated in the area of interest using `ee.Image.stratifiedSample()`. The sample size are based on the sample size calculation result. The feature collection from this process only contain class ID for each land cover class. Class name will be added in the future update. The user are able to download the samples, therefore they can labeled the sample outside Luma GE. Current function only support shapefiles.

#### Related Function

`sample_size_calculator.generate_stratified_samples`: Generate the reference samples based sample size calculation  
`sample_size_calculator.export_samples_to_shp`: Export the generated samples

### 7.2.3 Accuracy Computation Workflow

1. The system verifies if the `.shp` of the validation map is the uploaded inside the `.zip` file.

#### ! Error Handling Notification

The system provides an error notification if the `.shp` file is not found inside the `.zip` file.

2. The system conducts geometry fixes on the uploaded validation map.

#### 💡 Related Function

`input_utils.shapefile_validator.validate_and_fix_geometry()`: to validate and fix geometries issues.

#### ! Error Handling Notification

An error message is displayed if the system failed to run `validate_and_fix_geometry()`.

3. The system converts the validation map from `.gdf` data into Earth Engine Feature Collection format

#### 💡 Related Function

`input_utils.shapefile_validator.convert_roi_gdf()`: to convert geo-dataframe into EE Feature Collection. The supported multi-geometries type for this conversions are `MultiPoint` and `MultiPolygon`.

#### ! Error Handling Notification

An error message is displayed if the system failed to run `convert_roi_gdf()`.

## 7.2.4 Starting the thematic accuracy assessment

1. The system performs a thematic accuracy assessment by validating inputs, sampling the classified LULC map at reference points.

#### 💡 Related Functions

`accuracy.thematic_accuracy.validate_assessment_inputs()`: checks whether the classified LULC map, validation map, LULC ID class field, and pixel size parameter meet the required conditions before the assessment runs. This function is used in `run_accuracy_assessment()`.

```

accuracy.thematic_accuracy._extract_confusion_matrix_data(): extracts
overall accuracy, kappa, per-class accuracies, and the confusion matrix
array from an ee.ConfusionMatrix object. This function is used in
run_accuracy_assessment().
accuracy.thematic_accuracy._calculate_confidence_interval(): Computes
the confidence interval for overall accuracy using a normal approximation
based on the number of correct and total samples. This function is used in
run_accuracy_assessment().
accuracy.thematic_accuracy._calculate_f1_scores(): Calculates the F1
score for each class using the producer's and user's accuracy values. This function
is used in run_accuracy_assessment().
accuracy.thematic_accuracy.run_accuracy_assessment(): Executes the full
thematic accuracy workflow, including validation, sampling (sampleRegions()),
metric extraction (errorMatrix()), confidence interval calculation, and compilation
of final results.

```

## 2. Thematic Accuracy Metrics

Several accuracy metrics is calculated to provide comprehensive report of the map's thematic quality.

- **Overall Accuracy**

Overall Accuracy (OA): Sum of the major diagonal (correctly classified pixels) divided by the total pixels in the entire confusion matrix

$$OA = \Sigma(n_{ii}) / N$$

- **Kappa Coefficient**

Kappa coefficient is statistical test generated from the error matrix. Kappa coefficient show how well the classification performed as compared to just randomly assigning values. Kappa value range from -1 to 1, in which the value of 0 indicate that the classification is no better than a random classification. The negative value indicate that the classification is worse than random classification. The value closer to 1 indicate that the classification is better than random classification.

$$= (Po - Pe) / (1 - Pe) \quad Po = \Sigma(n_{ii}) / N \quad Pe = \Sigma(n_{i+} \times n_{+i}) / N^2$$

Where:

Po = Observed Agreement (OA)

Pe = Expected agreement by chance

- **Producers's Accuracy (Recall/Sensitivity)**

One of the class level accuracy metric, which answer the question, “*What fraction of actual class was correctly mapped?*”

$$PA_i = n_{ii} / n_{i+}$$

Where:

$n_{ii}$  = correctly classified samples for class i  $n_{i+}$  = total reference samples for class i (row sum)

- **User's Accuracy (Precision)**

One of the class level error metric, which answer the question “*What fraction of predicted class i is actually class i?*”

$$UA_i = n_{ii} / n_{+i}$$

Where:

$n_{ii}$  = correctly classified samples for class i  $n_{+i}$  = total predicted samples for class i (column sum)

## 7.2.5 Spatial Distribution of Error

This function provide reference data flagging for visualizing the spatial error distribution, improving the error analysis by tagging each point with whether the classifier got it right or wrong. Currently, this function only works if the reference data unit is point (single pixel). If the reference data consist of polygons, this function will not work.

1. The system conduct checks if reference data geodataframe is available.
2. Add actual and predicted class to the geodataframe as key point in determining correctness of the map
3. Generate a pop-up html with comprehensive information regarding the status of the corresponding reference sites

### Related Function

`accuracy.validation_error_flag.classify_validation_points()`: core function to perform correctness flagging to the classification map

`accuracy.validation_error_flag.generate_popup_html()`: create a pop up html for visualizing the error flag

### **7.2.6 Reviewing the thematic accuracy result**

This sub-step does not involve any operations from Luma Geospatial Engine.

## 8 Post Classification Analysis

 Under development

This module's backend is not available yet.

## **Part III**

# **Luma Interface**

## Map Generation

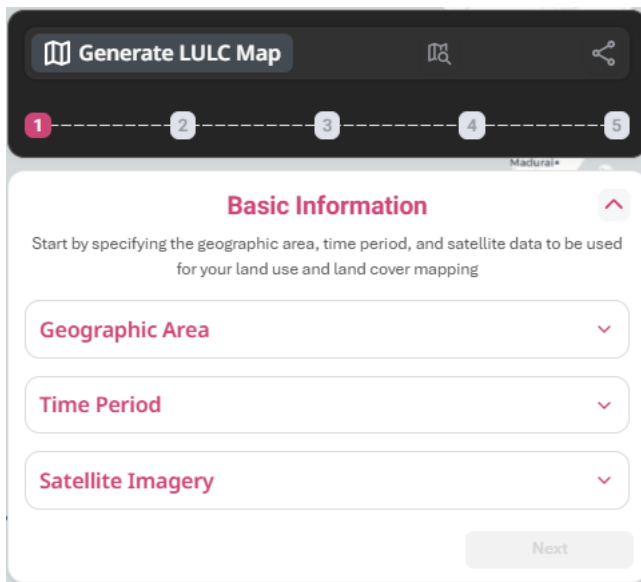
# Overview

In this workflow, the user produces a Land Use and Land Cover (LULC) map. Each step is presented as a panel on the left of the map canvas and progresses via the indicator at the top of the panel. The user cannot skip steps; each step unlocks the next only after the Geospatial Engine (GE) confirms the required output has been produced and saved.

# Step 1: Basic Information

## Overview

Provides the inputs needed before any geospatial computation: where to map, when, and from which satellite source.

The screenshot shows a mobile application interface for generating a Land Use and Land Cover (LULC) map. At the top, there is a dark header bar with a book icon, the text 'Generate LULC Map', a magnifying glass icon, and a share icon. Below the header is a progress indicator with five numbered steps (1-5), where step 1 is highlighted in red. The main content area is titled 'Basic Information' in red, with a small upward arrow icon to its right. Below the title, there is a subtitle: 'Start by specifying the geographic area, time period, and satellite data to be used for your land use and land cover mapping'. There are three input fields, each with a red label and a downward arrow icon: 'Geographic Area', 'Time Period', and 'Satellite Imagery'. At the bottom right of the form, there is a 'Next' button.

## Step 1.1: Geographic Area

### Overview

This is the first required input in the Map Generation workflow where the user specifies the geographic area they want to map.

Generate LULC Map

1

2

3

4

5

Thiruvananthapuram

Basic Information

Start by specifying the geographic area, time period, and satellite data to be used for your land use and land cover mapping

Geographic Area



Upload area

Define the area you want to map



Draw area

Draw a polygon on the displayed map to determine your area of interest



Select district/city

Choose an administrative district or city to use its boundary as your mapping area.

Time Period

Satellite Imagery

Next

### Step 1.1.1: Upload Area

#### Overview

The user uploads a polygon file to define the geographic boundary of the area they want to

105

map.

**Generate LULC Map**

1 2 3 4 5

**Geographic Area**

**Upload area**  
Upload an existing polygon file for the area you want to map

Drag and drop your polygon file here  
Accepted format: .zip (.shp, .shx, .dbf, .prj), .kml, .kmz

**Browse file**

**Spatial resolution:**  
Smallest dimension that can be distinguished in the area that you want to map.  
This reflects the level of detail of your map.

☐ 30 m x 30 m ☐ 100 m x 100 m  
☐ 500 m x 500 m ☐ 1 km x 1 km

**Reselect** **Confirm**

## Process

1. The user can upload their own AOI file in **.zip** format that includes all required shapefile components (**.shp**, **.shx**, **.dbf**, **.prj**).
2. The system checks on all required shapefile components that needs to be put in the **.zip** file. The system also supports uploads of **.kml** or **.kmz** files.

### **i** Success Notification

If all shapefile components pass validation, continue to Process 3

### **!** Error Handling Notification

If the shapefile components do not match, an error warning is displayed: 'No **.shp** file found in the **.zip** file.'

3. The system retrieves AOI file information.

**i** Success Notification

If all file information is correct, continue to Process 4

**!** Error Handling Notification

If the file information cannot be found, an error warning is displayed: 'AOI not found'

4. The system retrieves AOI file area size information.

**i** Success Notification

If the area size information is within the maximum area allowed, continue to Process 5

**!** Error Handling Notification

If the area size information is too large, an error warning is displayed: 'Draw area exceeds maximum limit'

5. The system validates the geometry of the AOI for both .shp and .kml using the **shapefile\_validator** functions.

**i** Success Notification

If validation is successful, continue to Process 6

**!** Error Handling Notification

If validation failed, an error warning is displayed: 'Geometry validation failed.'

6. The system returns the validated and processed shapefile.

**i** Success Notification

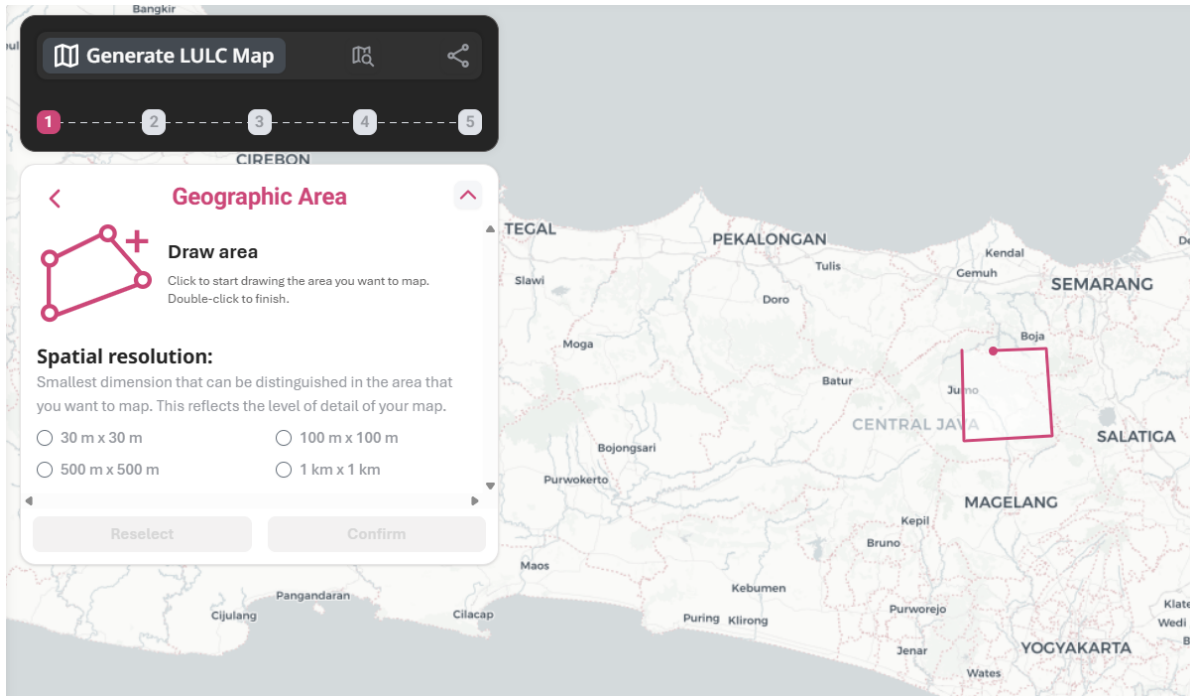
The system shows a confirmation when the AOI is loaded successfully by showing a purple box with 'Approximate size of your mapping area: X [number] Ha' displayed

7. If the user clicks on "Reselect Area", the uploaded file is deleted, and the user will be redirected to **?@sec-map-generation-basic-information**.

## Step 1.1.2: Draw Area

### Overview

The user draws their area of interest directly on the map by creating borders of area they want to map. Each click adds a new corner to the polygon, with a double-click to finish drawing and close the polygon. This is useful when the user does not have a prepared polygon file or wants to quickly define a simple area.



### Process

1. The user creates a polygon by clicking on the map and selects a spatial resolution from the available options.
2. The system retrieves the AOI area size information.

#### **i** Success Notification

The system shows a confirmation when the AOI is loaded successfully by showing a purple box with 'Approximate size of your mapping area: X [number] Ha' displayed

### ! Error Handling Notification

If the area size information is too large, an error warning is displayed: 'Draw area exceeds maximum limit'

3. If the user clicks on “Reselect Area”, the polygon is deleted, and the user will be redirected to [?@sec-map-generation-basic-information](#).

## Step 1.1.3: Administrative Boundaries

### Overview

The user selects the area they want to map based on a drop down list of Indonesian regency administrative boundaries. The system provides a preview of the selected boundary.

**Generate LULC Map**

1 2 3 4 5

Thiruvananthapuram

**Geographic Area**

**Select district/city**  
Choose an administrative district or city to use its boundary as your mapping area.

Select district/city

**Spatial resolution:**  
The smallest dimension that can be distinguished in the area that you want to map. This reflects the level of detail of your map.

☐ 30 m x 30 m ☐ 100 m x 100 m  
☐ 500 m x 500 m ☐ 1 km x 1 km

Reselect Confirm

### Process

1. The user selects an administrative boundary from the dropdown and a spatial resolution from the available options.
2. The system retrieves the AOI information from the selected administrative boundary selections. The system runs the related functions in Section 1.1.2 and provides a preview of the selected boundary.

## Step 1.2: Time Period

### Overview

The user defines the time period for the land use and land cover map. User chooses the range of period of satellite imagery in a single year format or a custom date range format.

The screenshot shows a web application interface for generating a Land Use and Land Cover (LULC) map. At the top, there is a dark header bar with a 'Generate LULC Map' button, a search icon, and a share icon. Below the header is a progress bar with five steps, where step 1 is highlighted. The main content area is titled 'Basic Information' and includes instructions: 'Start by specifying the geographic area, time period, and satellite data to be used for your land use and land cover mapping'. There are three main sections: 'Geographic Area' with a dropdown arrow, 'Time Period' with a dropdown arrow and a 'Set time period' button, and 'Satellite Imagery' with a dropdown arrow. A 'Next' button is located at the bottom right of the form.

### Process

1. The user chooses a time interval selection “By year” and a year (1972–current year).
2. The system retrieves the selected year and the options for the Satellite Imagery (?@sec-map-generation-satellite-imagery) are limited to the available sensor images depending on the selected year.

#### **i** Success Notification

The system shows a confirmation by showing a purple box with a summary of: time interval, period, satellite imagery date range selected

#### **!** Error Handling Notification

If time interval nor year is not selected, the options on Satellite Imagery (**?@sec-map-generation-satellite-imagery**) are not available

3. The user chooses a time interval selection "Custom date range" along with the start and end dates.
4. The system provides a preview of calendar for user to choose the start and end dates.

#### **i** Success Notification

The system shows a confirmation by showing a purple box with a summary of: time interval, period, satellite imagery date range selected

#### **!** Error Handling Notification

If time interval nor date range is not selected, the options on Satellite Imagery (**?@sec-map-generation-satellite-imagery**) are not available

## **Step 1.3: Satellite Imagery**

### **Overview**

The user chooses a satellite imagery source and sets the maximum cloud coverage. The system retrieves all available scenes, removes cloudy pixels, and combines the rest into a single cloud-free composite image. This composite is the base image for all classification steps. Three visualisation modes are generated: True Colour (RGB), False Colour Infrared, and Land/Water.

**Generate LULC Map**

1 2 3 4 5

### Basic Information

Start by specifying the geographic area, time period, and satellite data to be used for your land use and land cover mapping

**Geographic Area**

**Time Period**

**Satellite Imagery**

The most recent satellite imagery source for your selected period is chosen as default source, with maximum cloudy area set at 30%. Make adjustments only if you need a more specialized analysis using your own classification scheme.

Satellite imagery source

Landsat 8

Maximum cloudy area

0 10 20 30 40 50 60 70 80 90 100

Set satellite imagery

Next

## Process

1. The system checks the selected year in `?@sec-map-generation-time-period` and shows the available sensors based on the selected year.
2. The user sets a preferred cloud cover threshold, if none chosen, the default value of 30% is used.
3. The system performs pre-processing functions in Section 1.3.1.

### **i** Success Notification

When all information is collected, continue to Process 4

#### ! Error Handling Notification

The system provides an error notification if no satellite images were found based on the user's parameter. The user is prompted to change the cloud cover threshold or change the time period.

4. The system returns the collected satellite image information.

#### i Success Notification

The user's input summary is displayed in a purple box, showing:

- Mapping area: approximate size of your mapping area
- Time period selected: time interval, selected period, satellite input date range
- Satellite imagery: satellite imagery source, maximum cloudy area

Two buttons appear:

- "Change Input"; continue to Process 5
- "View Satellite Imagery"; continue to Process 8

#### ! Error Handling Notification

The system provides an error notification if no satellite images were found based on the user's parameter. The user is prompted to change the cloud cover threshold or change the time period.

5. If the user clicks on the "Change Input" button, "Edit" (pencil icon) buttons appear next to each of the input summary's heading. Once clicked, a pop up window appeared for user to choose:
  - "Change Input", continue to Process 6, or
  - "Discard Changes", continue to Process 7.
6. If the user clicks on the "Change Input" button in the pop up window, all inputs are being reset, back to the beginning of **?@sec-map-generation-basic-information**.
7. If the user clicks on the "Discard Changes" button in the pop up window, the panel still displays the user's input summary. "Cancel Change" button appear, while "View Satellite Imagery" is disabled. Once the "Cancel Change" button is clicked, back to Process 4.
8. If the user clicks on the "View Satellite Imagery" button, the system returns the collected satellite image information.

#### **i** Success Notification

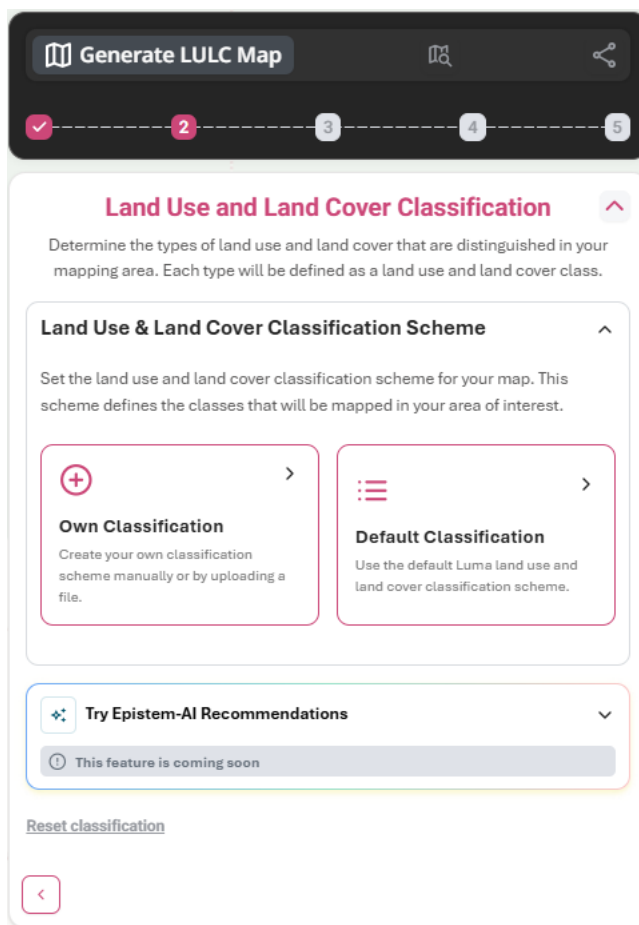
- A table of Generated Composite of Imagery Obtained appeared under the User's Input Summary along with Cloud Coverage Statistic.
- Composite of satellite imagery appears on the base map, with default setting of true colour (RGB).
- Layers option appears on the bottom right corner of the screen, includes:
  - Mapping area; with polygon toggle
  - Composite of satellite imagery; with true colour (RGB), false colour infrared (NIR/Red/Green), land/water (NIR/SWIR1/RED) toggle options
  - LULC class; still disabled in this step
- User can then download the composite, leading to Process 9.
- “Next” button appears, which leads to **?@sec-map-generation-lulc-classification**.

9. If the user downloads composite, a .zip file containing satellite image composite is downloaded to user's local device.

## Step 2: Land Use and Land Cover Classification

### Overview

The user defines the land use and land cover classes that will be mapped. There are two main options: using the built-in Luma default classification scheme or uploading a custom classification scheme. The selected classes determine which training data will be used in the next step. The Epistem-AI feature will also be available here.



## **Step 2.1: Land Use & Land Cover Classification Scheme**

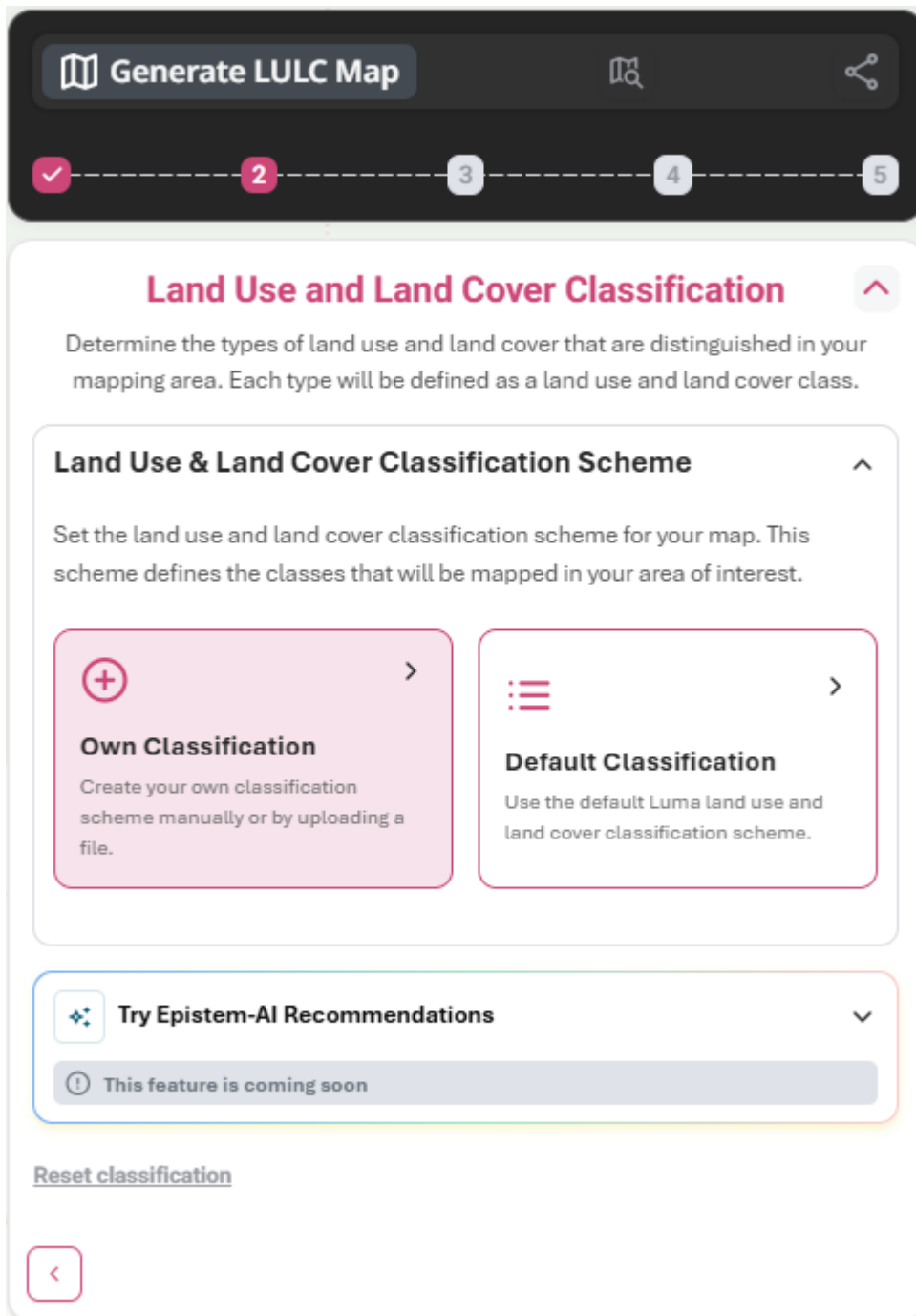
### **Overview**

Luma supports user-defined classification schemes and default classification or templates for commonly used classification schemes.

### **Step 2.1.1: Own Classification**

#### **Overview**

The user creates their own classification scheme manually or by uploading a file.



### Step 2.1.1.1: Manual Input

#### Overview

The user inputs the land use and land cover classes they want to map. They need to type a class name and optionally pick a colour for each class.

## Process

1. The user enters LULC class name and pick a colour out of 28 colour options.
2. The system stores each class entry individually as it is added.
3. The system validates class ID, name, and checks for uniqueness with the functions in Section 2.1.2.

### **i** Success Notification

- All added classes are displayed in the class recorded table for preview
- Upload Classes option still available; if clicked, continue to **?@sec-map-generation-upload-classes** Process 1
- Confirm LULC Class enabled; once clicked, all inputs are locked and a summary table is previewed, along with three options:
  - “Edit” (pencil icon); continue to Process 4
  - “Download LULC classes (CSV)”; a **.csv** file containing the inputted classification scheme is downloaded to user’s local device.
  - “Next”; continue to Process 5

**! Error Handling Notification**

The system returns a validity flag with an error message if invalid

4. If the user edits or deletes the classes in the summary table preview, the system stores each class entry individually as it is added and runs the functions in Section 2.1.2.

**i Success Notification**

- Each class row can be edited individually
- New classes can be added with Add Class button
- If all classes are deleted, back to the beginning of **?@sec-map-generation-lulc-classification**
- “Done” button appears; if clicked:
  - “Next” button appears; continue to Process 5
  - “Download LULC classes (CSV)” appears; a .csv file containing the inputted classification scheme is downloaded to user’s local device

**! Error Handling Notification**

The system returns a validity flag with an error message if invalid

5. If the user clicks on the “Next” button, the user will be directed to **?@sec-map-generation-sample-data**.
6. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-satellite-imagery** Process 8 is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-lulc-classification**.

### Step 2.1.1.2: Upload Classes

#### Overview

The user downloads a classification template .csv file and uses it to list the land use and land cover classes they want to map. After filling the template, they need to upload the file to be used as the custom classification scheme.

## Process

1. The user downloads the `.csv` file containing the classification scheme information template into the user's local device.
2. The user uploads the `.csv` file containing the classification scheme information.
3. The system stores the file uploaded and runs the functions in Section 2.1.1.

### Success Notification

If the classification scheme file successfully uploaded:

- A purple box indicating the uploaded file with its file size appears
- A summary table of recorded classes with its ID number and colour class appears
- Manual Input option still available; if clicked, continue to **?@sec-map-generation-manual-input** Process 3
- Confirm LULC Class enabled, if clicked, a summary table is previewed, along with three options:
  - “Edit” (pencil icon); continue to Process 4
  - “Download LUCL classes (CSV)”; a `.csv` file containing the inputted classification scheme is downloaded to user's local device.
  - “Next”; continue to Process 5

**! Error Handling Notification**

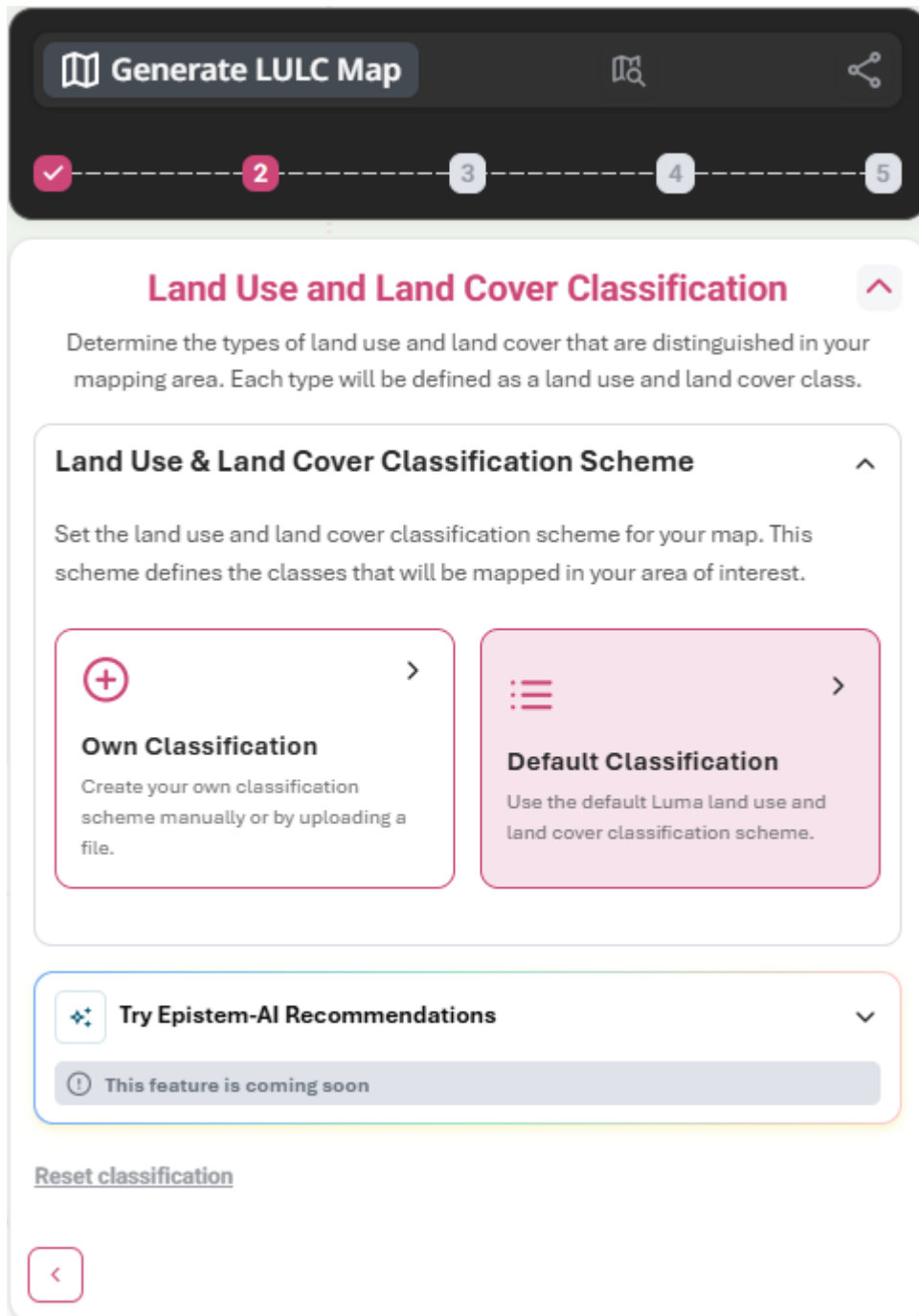
The system returns a validity flag with an error message if invalid

4. If the user clicks on the “Edit” button, each class row can be edited individually. The uploaded file can be deleted, if clicked, all inputs are being reset, back to the beginning of **?@sec-map-generation-lulc-classification**.
5. If the user clicks on the “Next” button, the user will be directed to **?@sec-map-generation-sample-data**.
6. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-satellite-imagery** Process 8 is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-lulc-classification**.

### **Step 2.1.2: Default Classification**

#### **Overview**

The user activates class options from default Luma land use and land cover classification scheme. The user is given the option to choose only a specific subset of the default classes, rather than being required to utilize the entire set.



## Process

1. The user selects subsets of the default classes.
2. The classification table is being saved to the system.

3. The system loads the selected classes from a dataset by running functions in Section 2.1.3.

#### Success Notification

- A summary table of recorded classes with its ID number and colour class appears
  - Two buttons appear:
    - “Reset classification”; continue to Process 4
    - “Next”; continue to Process 5
4. If the user clicks on “Reset classification”, All inputs are being reset, back to the beginning of **?@sec-map-generation-lulc-classification**.
  5. If the user clicks on the “Next” button, the user will be directed to **?@sec-map-generation-sample-data**.
  6. If the user clicks on the “Edit” button, all inputs are being reset, back to the beginning of **?@sec-map-generation-lulc-classification**.
  7. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-satellite-imagery** Process 8 is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-lulc-classification**.

## Step 2.2: Try Epistem-AI Recommendations

### Overview

This feature is coming soon.

 **Try Epistem-AI Recommendations** 

 This feature is coming soon

I need to map forested areas with minimal human activity, only small-scale traditional farming, and no large infrastructure nearby.

Maximum 500 characters 

Generate

## Step 3: Mapping Sample Data

### Overview

The user prepares the sample data for land use and land cover classification in their mapping area. The user can upload existing data or create samples directly in Luma through on-screen sampling. In this step the Sample Data Summary selected in the previous step is also shown. The spectral separability of LULC classes is also being assessed in this step using sample data and near-cloud-free satellite imagery. Pairwise class separability is quantified using the Transformed Divergence (TD) method based on pixel-level spectral statistics, using only the spectral data from **?@sec-map-generation-basic-information**.

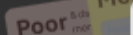
### Step 3.1: Upload Sample Data

#### Overview

The user uploads their existing sample data in **.zip** file format.

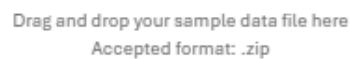
[See More...](#)

Check Score 



## On Screen Sampling

💡 A minimum of 20 samples per class is recommended but not enforced.



[Browse file](#)

^

| No. | LULC class | Color Class | Number of samples | ⑧ |
|-----|------------|-------------|-------------------|---|
|-----|------------|-------------|-------------------|---|

## Process

1. The user uploads the **.zip** file containing the sample data.
2. The system checks if there are any **.shp** file inside the **.zip** file by running the functions in Section 3.2.1.
3. The system returns the preview of the uploaded sample data on the map canvas and table of the uploaded sample data.

### Success Notification

- A purple box indicating the uploaded file with its file size appears with a delete button, if clicked, continue to the beginning of **?@sec-map-generation-sample-data**
- A summary table of Sample Data Summary with ID number, LULC class name, colour class, number of samples, and buttons to view each class appears

### Error Handling Notification

The system provides an error notification if the system fails to conduct the sample data verification, along with the specific message at which step the validation failed

4. The user clicks on the “Check Score” button to know their sample data quality.
5. The system provides a selection for Spatial Resolution (m) and Maximum Pixels Per Class. The default value is 100 m and 5000 pixels.
6. The system conducts the separability analysis by running the functions in Chapter 4.

### Success Notification

- Training Data Quality result appears, with its category (good, weak, or poor) for each class, and highlights which LULC classes require improvement. A data frame containing details of the pairwise separability is also displayed.
- The user can choose several paths:
  - Delete the uploaded file; all input in **?@sec-map-generation-upload-sample-data** is deleted and continue to the beginning of **?@sec-map-generation-sample-data**
  - On Screen Sampling:
    - \* Single Point Input; continue to **?@sec-map-generation-single-point**
    - \* Bulk Point Input; continue to **?@sec-map-generation-bulk-point**
  - Next; continue to **?@sec-map-generation-parameters**

7. If the user clicks on the “Recheck Score” button, continue to Process 5 and 6.
8. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-lulc-classification** summary table of recorded classes is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-sample-data**.

## Step 3.2: On Screen Sampling

### Overview

The user adds points or draws polygons directly on the interactive displayed map for all land use and land cover classes in their mapping area.

Generate LULC Map

✓

✓

3

4

5

## Mapping Sample Data

Sample data is a small number of locations where presence of the land use and land cover classes that you specify can be determined. Sample data...

[See More...](#)

🕒 Know your training data score

Generate the separability analysis to see how well your sample classes can be distinguished from each other.

Check Score

▼

Upload Sample Data

On Screen Sampling

Add points or draw polygons directly on the displayed map for all land use and land cover classes in your mapping area.

💡 A minimum of 20 samples per class is recommended but not enforced.

Add point

Draw area

Add points on the displayed map and input their LULC class information.

📍+

Single Point Input

Add a point and assign its corresponding LULC class one by one.

📍📍+

Bulk Point Input

Add multiple points (up to 10 points) at once and assign the same LULC class to all selected points.

◀

Next →

128

### Step 3.2.1: Add Point

#### Overview

The user adds points on the interactive displayed map and input their land use and land cover class information.

#### Step 3.2.1.1: Single Point Input

#### Overview

The user adds a point and assigns its corresponding land use and land cover class one by one from the drop down list.

Upload Sample Data

On Screen Sampling

Add points or draw polygons directly on the displayed map for all land use and land cover classes in your mapping area.

 A minimum of 20 samples per class is recommended but not enforced.

Add point

Draw area

Add points on the displayed map and input their LULC class information.



**Single Point Input**  
Add a point and assign its corresponding LULC class one by one.



**Bulk Point Input**  
Add multiple points (up to 10 points) at once and assign the same LULC class to all selected points.

Start sampling

#### Process

1. The user assigns a sample point on the map accordingly to the correct class from the drop down list.

2. The system shows a confirmation that the pinned points have been loaded to the on-screen canvas, along with the number of sample data on the Summary Table.

**i** Success Notification

- The number of samples in the Summary Table increased based on the user's inputs
- End Pointing button enabled; if clicked, continue to Process 3

3. If the user clicks on “End Pointing” button, the user is redirected to **?@sec-map-generation-oss** with Sample Data Summary table and “Training Data Score Check” button enabled.
4. The user clicks on the “Check Score” button to know their sample data quality.
5. The system provides a selection for Spatial Resolution (m) and Maximum Pixels Per Class. The default value is 100 m and 5000 pixels.
6. The system conducts the separability analysis by running the functions in Chapter 4.

**i** Success Notification

- Training Data Quality result appears, with its category (good, weak, or poor) for each class, and highlights which LULC classes require improvement. A data frame containing details of the pairwise separability is also displayed.
- The user can choose several paths:
  - Delete the uploaded file; all input in **?@sec-map-generation-upload-sample-data** is deleted and continue to the beginning of **?@sec-map-generation-sample-data**
  - On Screen Sampling:
    - \* Single Point Input; continue to **?@sec-map-generation-single-point**
    - \* Bulk Point Input; continue to **?@sec-map-generation-bulk-point**
  - Upload Sample Data; continue to **?@sec-map-generation-upload-sample-data**
  - Next; continue to **?@sec-map-generation-parameters**

7. If the user clicks on the “Recheck Score” button, continue to Process 5 and 6.
8. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-lulc-classification** summary table of recorded classes is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-sample-data**.

### Step 3.2.1.2: Bulk Point Input

#### Overview

The user adds multiple points (up to 10 points) at once and assigns the same land use and land cover class to all selected points.

The screenshot shows a web interface for 'On Screen Sampling'. At the top, there are two buttons: 'Upload Sample Data' (grey) and 'On Screen Sampling' (pink). Below them is a text instruction: 'Add points or draw polygons directly on the displayed map for all land use and land cover classes in your mapping area.' A progress bar shows 'Add point' as the active step, with 'Draw area' as the next step. Below the progress bar, a text instruction reads: 'Add points on the displayed map and input their LULC class information.' There are two input options: 'Single Point Input' (unselected, with a single point icon) and 'Bulk Point Input' (selected, with a multiple points icon). The 'Bulk Point Input' section includes a text instruction: 'Add multiple points (up to 10 points) at once and assign the same LULC class to all selected points.' Below this is a 'LULC class' label and a dropdown menu with the text 'Select LULC class'. At the bottom of the interface is a 'Start sampling' button.

#### Process

1. The user chooses a class from the drop down list and assigns sample points on the map accordingly to the correct class.
2. The system shows a confirmation that the pinned points have been loaded to the on-screen canvas, along with the number of sample data on the Summary Table.

#### Success Notification

- The number of samples in the Summary Table increased based on the user's inputs
- End Pointing button enabled; if clicked, continue to Process 3

3. If the user clicks on “End Pointing” button, the user is redirected to **?@sec-map-generation-oss** with Sample Data Summary table and “Training Data Score Check” button enabled.
4. The user clicks on the “Check Score” button to know their sample data quality.

5. The system provides a selection for Spatial Resolution (m) and Maximum Pixels Per Class. The default value is 100 m and 5000 pixels.
6. The system conducts the separability analysis by running the functions in Chapter 4.

#### **i** Success Notification

- Training Data Quality result appears, with its category (good, weak, or poor) for each class, and highlights which LULC classes require improvement. A data frame containing details of the pairwise separability is also displayed.
- The user can choose several paths:
  - Delete the uploaded file; all input in **?@sec-map-generation-upload-sample-data** is deleted and continue to the beginning of **?@sec-map-generation-sample-data**
  - On Screen Sampling:
    - \* Single Point Input; continue to **?@sec-map-generation-single-point**
    - \* Bulk Point Input; continue to **?@sec-map-generation-bulk-point**
  - Upload Sample Data; continue to **?@sec-map-generation-upload-sample-data**
  - Next; continue to **?@sec-map-generation-parameters**

7. If the user clicks on the “Recheck Score” button, continue to Process 5 and 6.
8. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-lulc-classification** summary table of recorded classes is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-sample-data**.

## Step 3.2.2: Draw Area

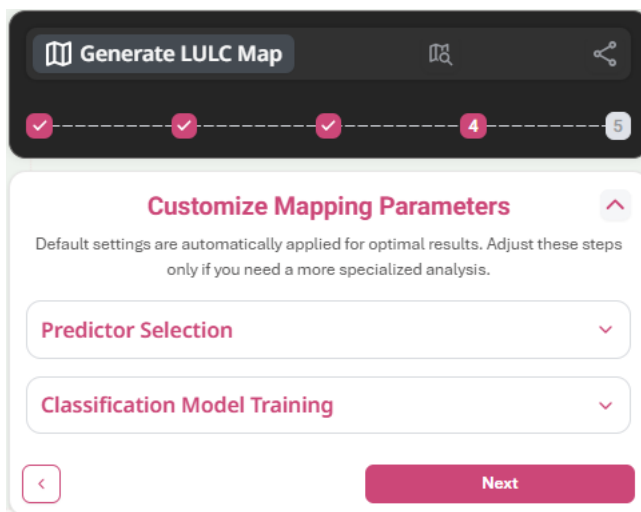
### Overview

This feature is coming soon.

## Step 4: Customize Mapping Parameters

### Overview

The user can choose a set of predictors that optimize the discrimination of LULC classes in the classification model. Predictors are generated from three principal categories: spectral transformations (indices), terrain-based metrics, and distance-based metrics, capturing complementary spectral, topographic, and spatial information.



### Step 4.1: Predictor Selection

#### Overview

The user is prompted to select the predictors currently supported by Luma Geospatial Engine by using check boxes or upload their own predictor data in .zip file.

#### Step 4.2.1: Default Predictors

##### Overview

The user selects the predictors currently supported by Luma Geospatial Engine by using check boxes.

### Predictor Selection

Choose additional predictors, such as spectral indices, topography, or upload your own predictor data to better distinguish LULC classes.

Vegetation

☐ NDVI  
Normalized Difference Vegetation Index

☐ EVI  
Enhanced Vegetation Index

Water & humidity

☐ MNDWI  
Modified Normalized Difference Water Index

Land & built-up

☐ Elevation  
Shuttle Radar Topography Mission (SRTM) elevation

☐ Slope  
Shuttle Radar Topography Mission (SRTM) slope

☐ NDBI  
Normalized Difference Built-Up Index

## Process

1. The user selects the predictors from the list.
2. The system retrieves the predictors based on the user's selection and runs the functions in [Section 5.1](#).

#### **i** Success Notification

- “Next” button leads to Summary of LULC List Parameters showing the predictors selected and Random Forest Variable in a purple box.
- Two buttons appear:
  - “Change Input”; continue to Process 3
  - “Next”; continue to Process 4

3. The user clicks on “Change Input”.

#### **i** Success Notification

- “Change Input”; all inputs being reset and back to the beginning of **?@sec-map-generation-parameters**
- “Discard Changes”; stay in Process 3 with “Cancel Change” button activated

4. If the user clicks on “Next” button, the system generates map with all input from Step 1-4.

#### **i** Success Notification

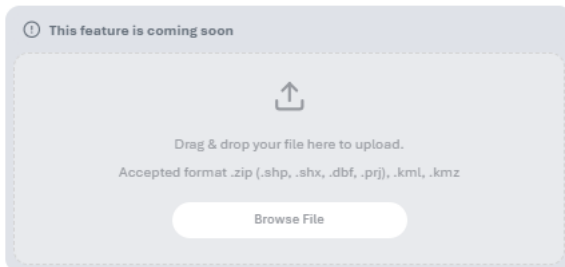
A pop up window appears with User’s Input Summary and Summary of LULC Class Compositions in purple boxes, with a “Generate Map” button. Continue to **?@sec-map-generation-final**.

5. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-sample-data** Sample Data Summary is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-parameters**.

## Step 4.2.2: Upload Predictors

### Overview

This feature is coming soon.



## Step 4.2: Classification Model Training

### Overview

The user is given the option to adjust the classification model by adjusting the value of the Random Forest variable to classify the satellite images into land use and land cover classes using training data. The default value of the Random Forest parameters are 150 number of trees and 1 minimum leaf population. The user is also prompted to define the proportion to split the sample data into training and validation dataset. The default value of the dataset proportion is 70% for training data and 30% for validation data.

### Classification Model Training

Random Forest is used to classify satellite images into LULC classes using training data. It works by combining many decision trees, which each make a prediction for every pixel. The final result is based on the majority vote from all trees, making the classification more accurate and reliable.

Number of trees

150

Fill in a number between 10 and 500

Minimum leaf population

1

Fill in a number between 1 and 50

#### Training / testing split

0102030405060708090100

Training70%

Testing30%

Set Variable

### Process

1. The user sets the Random Forest parameters and Training/Testing Split.
2. The system verifies the inputs and runs the process to create the classification model in [Section 6.3](#) and splits the sample data with functions in [Section 6.2.1](#).

**i** Success Notification

“Set Variable” button leads to Summary of LULC List Parameters showing the predictors selected and Random Forest Variable in a purple box. Two buttons appear:

- “Change Input”; continue to Process 3
- “Next”; continue to Process 4

3. The user clicks on “Change Input”.

**i** Success Notification

“Edit” (pencil icon) appears next to each parameter; if clicked, a pop up window appears asking if user wants to:

- “Change Input”; all inputs being reset and back to the beginning of **?@sec-map-generation-parameters**
- “Discard Changes”; stay in Process 3 with “Cancel Change” button activated

4. If the user clicks on “Next” button, the system generates map with all input from Step 1-4.

**i** Success Notification

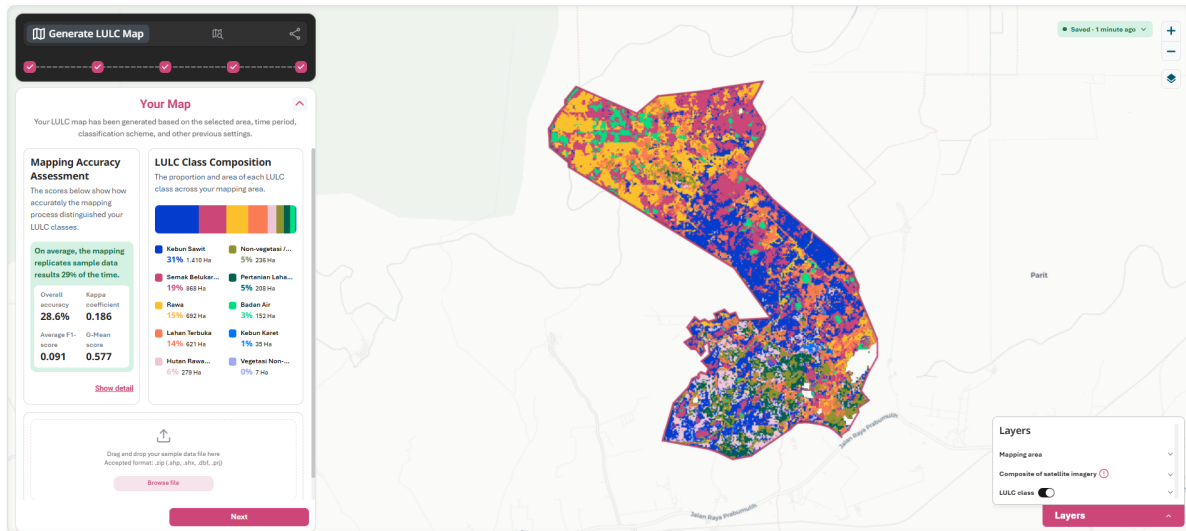
A pop up window appears with User’s Input Summary and Summary of LULC Class Compositions in purple boxes, with a “Generate Map” button. Continue to **?@sec-map-generation-final**.

5. If the user clicks on the “Back” (chevron) button, the panel of **?@sec-map-generation-sample-data** Sample Data Summary is being shown. The user has to click the “Next” button to go back to the last edited panel in **?@sec-map-generation-parameters**.

# Step 5: Map Generation

## Overview

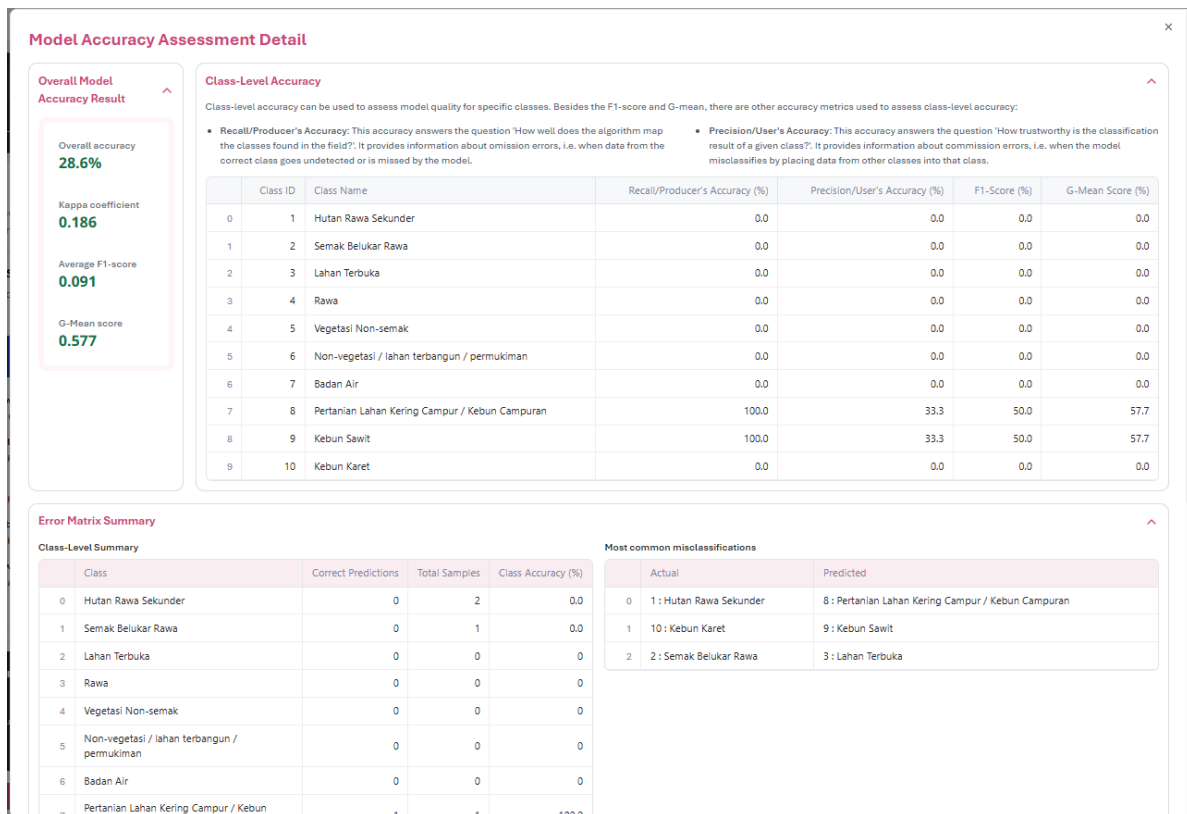
The user is shown the map generated on the map canvas with the classes in different colours, and the panel is showing model accuracy results, LULC class composition, and map's thematic accuracy assessment process.



## Step 5.1: Model Accuracy Assessment

### Overview

The user is shown how accurately the mapping process distinguished their land use and land cover classes.



## Process

1. The user clicks on “Show Detail”.
2. The system retrieves the mapping accuracy assessment result.

### Success Notification

A pop up window shows:

- Overall Model Accuracy Result
  - Overall accuracy
  - Kappa coefficient
  - Average F1-score
  - G-mean score
- Class Level Accuracy
  - Recall/Producer's accuracy
  - Precision/User's accuracy

- Error Matrix Summary
  - Class-level summary
  - Most common misclassifications
- Confusion Matrix

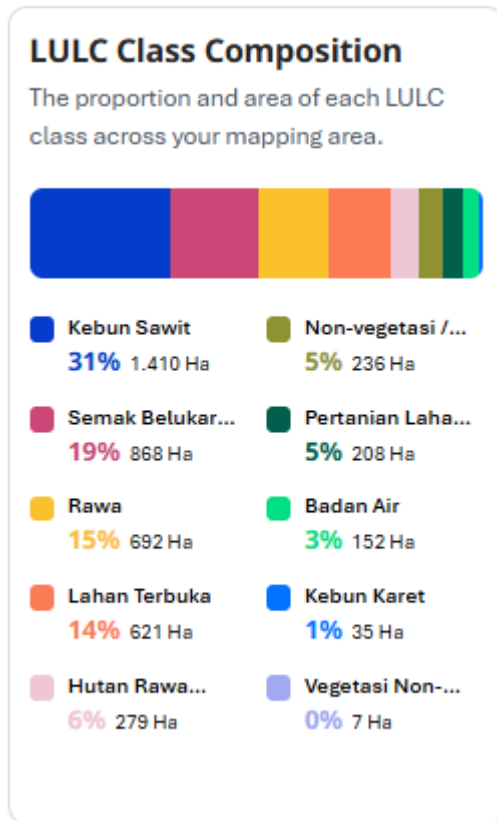
Two buttons appear at the bottom:

- “Download Raw Data”; .csv file containing the raw data is downloaded to user’s local device
- “Download Chart”; .png file of the confusion matrix is downloaded to user’s local device

## Step 5.2: LULC Class Composition

### Overview

The user can visualize the LULC classification result on the map canvas along with the training data layer and is shown the proportion of each LULC class.



## Step 5.3: Thematic Accuracy Assessment

### Overview

The user uploads an independent ground reference data in .zip file to perform a comprehensive thematic accuracy assessment of the land cover map.

## Thematic Map Accuracy Assessment

The thematic accuracy assessment measures how accurate the generated LULC map is using independent validation data.



Drag & drop your file here to upload.

Accepted format .zip (.shp, .shx, .dbf, .prj), .kml, .kmz

Browse File

### Process

1. The user uploads .zip file containing the ground reference data.
2. The system verifies if the .shp of the validation map is the uploaded inside the .zip file and runs the functions in Section 7.2.

#### Success Notification

A pink box showing the Thematic Accuracy Result appears with overall prediction, Kappa coefficient, number of points correct and wrong. Two actions can be chosen:

- “Reupload validation data”
- “Show detail”; continue to Process 4

#### Error Handling Notification

The system shows an error message if the system fails to run the thematic accuracy assessment

3. If the user clicks on “Reupload validation data”, continue to the beginning of **?@sec-map-generation-final**.
4. If the user clicks on “Show detail”, the system previews the comprehensive report of the map’s thematic quality.

#### **i** Success Notification

A pop up window shows:

- Overall Model Accuracy Result
- Overall accuracy
- Kappa coefficient
- Average F1-score
- G-mean score
- Class Level Accuracy
- Recall/Producer's accuracy
- Precision/User's accuracy
- Error Matrix Summary
  - Class-level summary
  - Most common misclassifications
- Confusion Matrix

Two buttons appear at the bottom:

- “Download Raw Data”; .csv file containing the raw data is downloaded to user's local device.
- “Download Chart”; .png file of the confusion matrix is downloaded to user's local device.

5. If the user clicks on the “Next” button.

#### **i** Success Notification

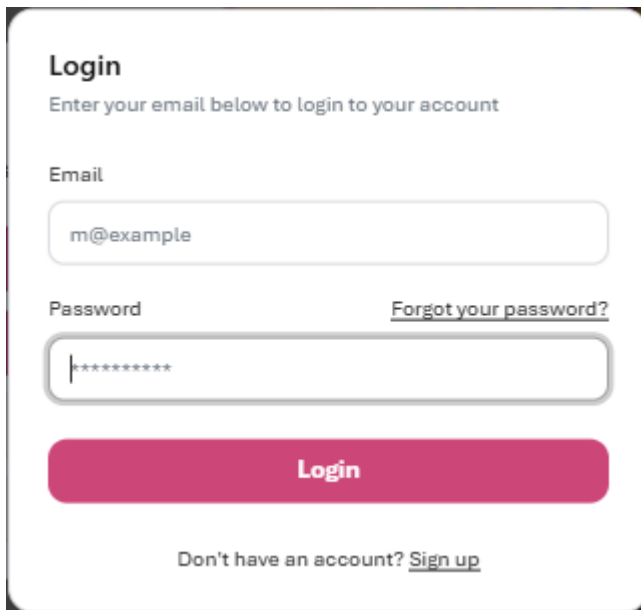
A pop up window shows several options:

- “Download map”; user is prompted to log in/sign up, continue to Feature 2
- “Share map”; user is prompted to log in/sign up, continue to Feature 3
- “Save to my account” (currently disabled)
- “Improve accuracy”

# Additional Features

## Overview

The user can save their land use and land cover mapping progress and share their mapping results by logging in/signing up to an Epistem account. Once logged in, user account is activated on the upper right corner of the screen with the user's initial or profile picture.

A login form with a white background and rounded corners. At the top, the word "Login" is in bold. Below it, a subtitle reads "Enter your email below to login to your account". There are two input fields: "Email" containing "m@example" and "Password" containing "\*\*\*\*\*". To the right of the password field is a link "Forgot your password?". Below the inputs is a large pink "Login" button. At the bottom, there is a link "Don't have an account? Sign up".

**Login**

Enter your email below to login to your account

Email

m@example

Password [Forgot your password?](#)

\*\*\*\*\*

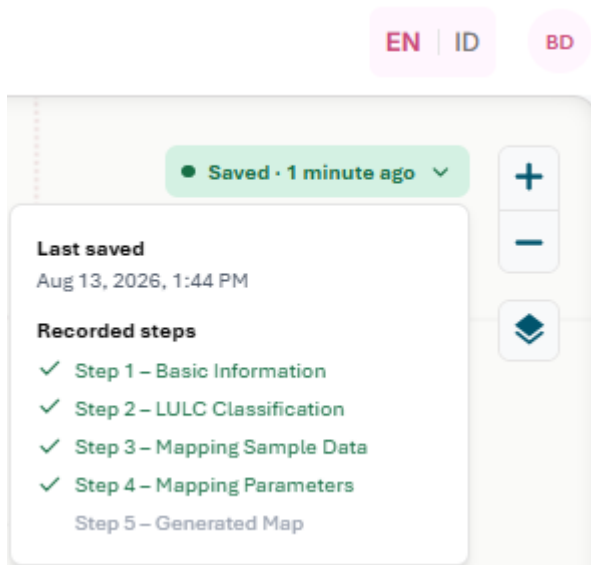
**Login**

Don't have an account? [Sign up](#)

## Feature 1: Save Progress

### Overview

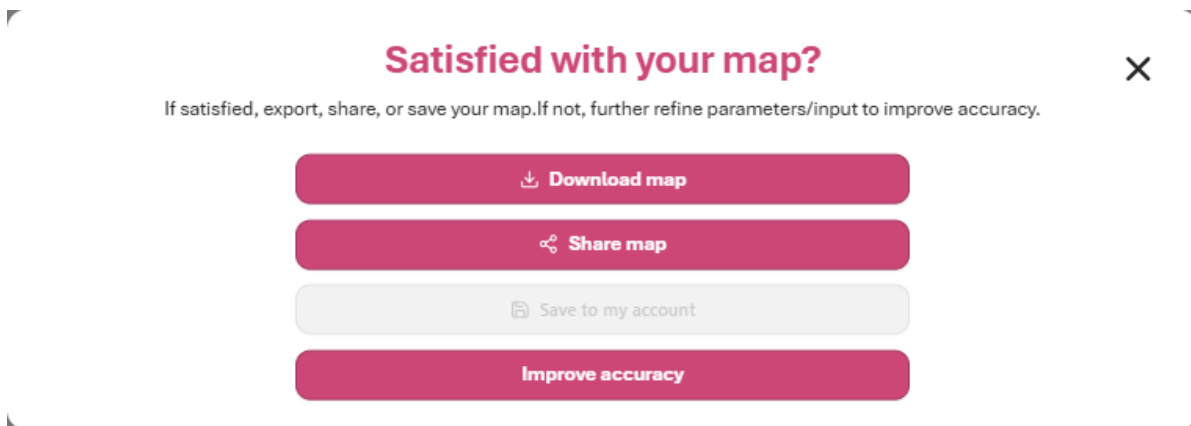
Once logged in, the user can save their land use and land cover mapping progress and view which steps they have done and where they are currently in. This can be viewed on the upper right corner of the map canvas. The session is stored for up to 24 hours within the database.



## Feature 2: Download Map

### Overview

Once logged in, the user can download their generated map.



### Process

1. The user downloads their map.
2. The system sends an email containing the exported LULC map with a download link.

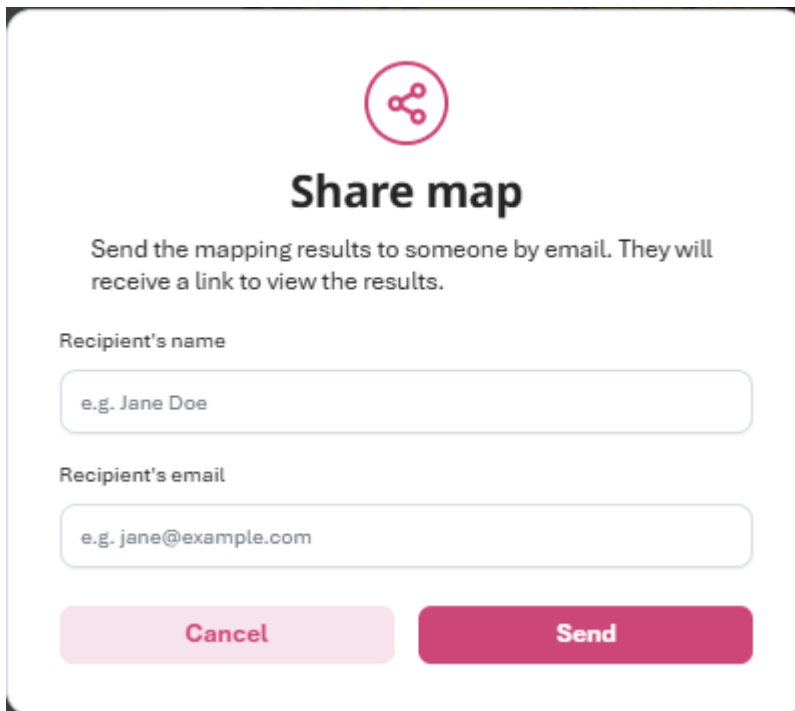
#### Success Notification

The user receives an email from [webdev@epistem.io](mailto:webdev@epistem.io) containing a download link to the GeoTIFF of the exported map

## Feature 3: Share Map

### Overview

Once logged in, the user can share their generated map to another person with an email address.



A modal dialog box titled "Share map" with a share icon (three nodes connected by lines) above the title. Below the title, a subtitle reads: "Send the mapping results to someone by email. They will receive a link to view the results." There are two input fields: "Recipient's name" with placeholder text "e.g. Jane Doe" and "Recipient's email" with placeholder text "e.g. jane@example.com". At the bottom, there are two buttons: "Cancel" (light pink) and "Send" (dark pink).

### Process

1. The user input recipient's name and email address.
2. The system sends and email to the recipient containing the exported LULC map with a download link.

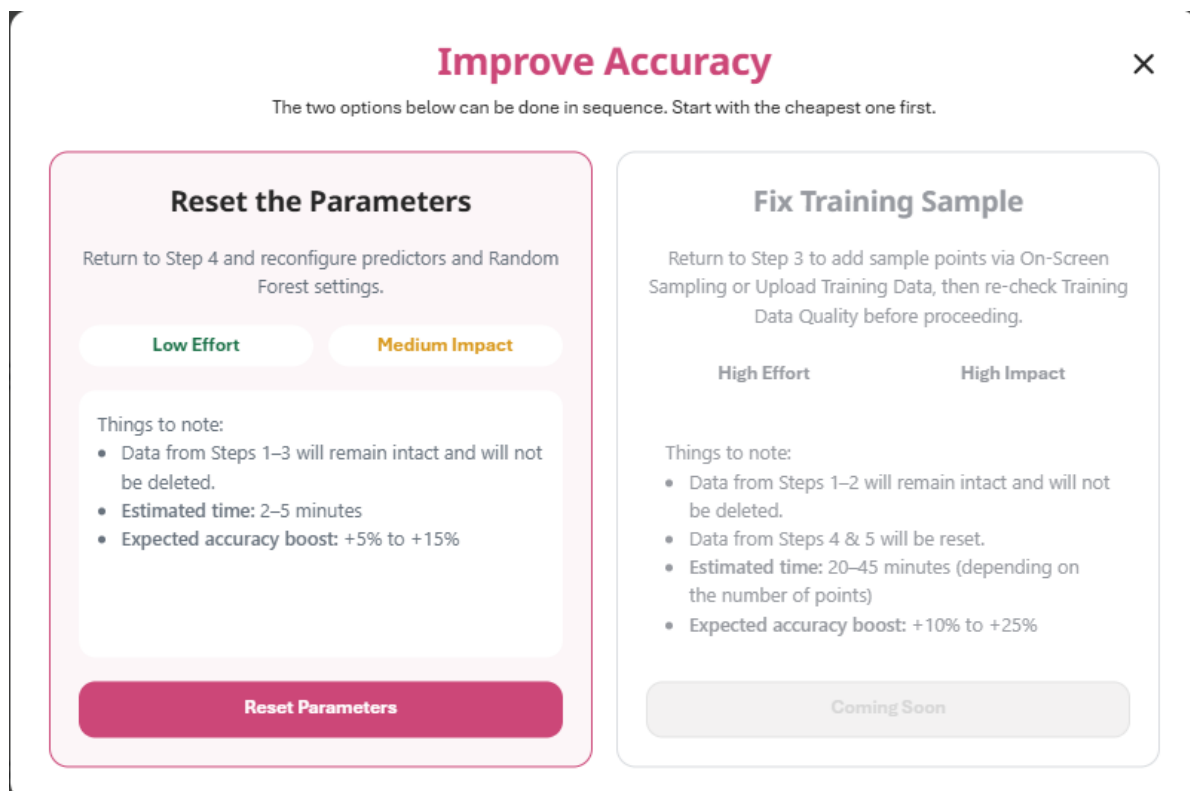
### Success Notification

The recipient receives an email from [webdev@epistem.io](mailto:webdev@epistem.io) containing a download link to the GeoTIFF of the exported map

## Feature 4: Improve Accuracy

### Overview

The user can improve the accuracy of their generated map in two ways: by resetting the parameters in `?@sec-map-generation-parameters`, or by adding or editing the sample data in `?@sec-map-generation-sample-data`.



### Process

1. The user clicks on "Improve Accuracy".

**i** Success Notification

A pop up window showing two methods of improving accuracy:

- "Reset Parameters"; continue to Process 2
- "Fix Training Data"; continue to Step 3 (coming soon)

2. If the user clicks on "Reset Parameters", the system retrieves information from Step 4.

**i** Success Notification

The panel returns to the beginning of **?@sec-map-generation-parameters**, with all selected predictors still checked, and displays a message explaining that the user is in Improvement Mode.

## **Change Analysis**

# Collaborative Mapping